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Designing for Relational Interaction in Horse-Human Communication

Exploring Haptic Feedback During Lunging

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Abstract

This thesis explores how a wearable haptic system can be designed to support horse-human interaction in lunging. Horse-human communication during lunging involves complex and embodied interaction that extends beyond measurable physical signals. While equestrian technologies often focus on performance and behavioural measurement, less attention has been given to supporting communication and awareness within ongoing interaction. This thesis explores how haptic feedback can be designed to support experiences of connection during horse-human interaction.

Using a Research-through-Design approach, the project combined observations, interviews, ideation and iterative prototyping. Findings show recurring interaction qualities associated with meaningful communication, including attentiveness, responsiveness, rhythmic consistency, willingness to engage and perceived connection. These insights informed the development of a wearable haptic prototype evaluated through Wizard-of-Oz testing.

The results suggested that haptic feedback may support awareness and reflection during interaction rather than functioning as a tool for control or performance measurement. The findings also indicated that experiences of connection are relational, subjective and developed over time rather than existing as fixed measurable states.

This thesis contributes to the field of Interaction Design through studying physical interaction that translates relational qualities into haptic feedback within multispecies contexts and highlights opportunities for technology to support mutual understanding in horse-human interaction.

Keywords: Horse-Human Interaction, Haptic Feedback, Animal-Computer Interaction Design, Embodied interaction, Research through Design

Declaration of AI Use

Artificial intelligence tools were used as supportive tools throughout the writing process of this thesis, AI assistance was used for generating suggestions related to text structure, wording and grammar.

AI was not used to generate research findings, conduct data analysis, interpret empirical material, or make research decisions. All analysis, interpretations, design decisions and conclusions were developed and evaluated by the author. The author takes full responsibility for the content and final presentation of this thesis.

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1. Introduction

1.1 The problem situation

Horse-Human interaction relies heavily on the interpretation of subtle, non-verbal signals. In practices such as lunging, a common training and assessment tool where the horse moves in a circle around a handler while attached to a long line, called lunge line. Through body language, voice cues and contact through the line, the handler guides the horse's speed, direction and posture. Lunging is used for exercise, training, rehabilitation and movement assessment, as it allows the handler to observe the horse's gait and behaviour without carrying a rider. The human continuously observes and responds to the horse's movement, posture, rhythm, breathing and body language to guide the interaction and assess the horse's physical and emotional state. This process is highly embodied and multisensory, involving vision, hearing and touch simultaneously. Research in equitation science has shown that horse-human communication depends on precise timing, body language, and the ability to read behavioural and physical cues, all of which influence welfare and performance (McGreevy & McLean, 2010).

At the same time, interpretation is complex and uncertain. Small changes in movement can indicate tension, imbalance, fatigue, discomfort or early signs of lameness, but these signals are often subtle and difficult to perceive consistently in real time. Studies of equine movement assessment show that human observation remains suggestive and dependent on experience which can create inconsistencies when identifying subtle asymmetries (Keegan et al., 2010). This makes perception during lunging a demanding activity where the human must continuously scan and interpret movement while remaining responsive to the ongoing communication between horse and human.

Animal–Computer Interaction (ACI) offers a framework for understanding how technology can be designed in ways that consider animals as active participants rather than passive objects in interaction systems (Manchini, 2011). Similarly, more-than-human design and multispecies design challenge human-centred perspectives by emphasizing relationality, attentiveness and care across species (Wakkary, Odom, Hauser, Hertz & Lin, 2015). These perspectives shift the focus from optimization and control toward awareness and mutual understanding.

Recent developments in equine technology, such as wearable sensors and gait analysis systems, provide new ways of measuring movement and detecting irregularities. These systems are increasingly used in training, rehabilitation and veterinary contexts (Serra Bragança et al., 2018). However, many of these technologies focus on performance measurement and diagnostic precision rather than how information is experiences and interpreted during the interaction itself. This creates an opportunity for interaction design to explore how technological systems can support perception without disrupting the existing horse–human relationship.

This project focuses specifically on lunging as situated interaction context. Through observations and field studies, it became clear that handlers rely on subtle and continuous sensory input, such as tension in the lunge line, rhythm of hoofbeats, breathing patterns and body posture, to interpret the horse's state and movement. Inspired by more-than-human design approaches and relational perspectives on human-animal interaction, this project

explores how physical and embedded interaction systems can support awareness of equine movement in real time, not to replace human interpretation, but to support noticing, interpretation and care.

The project is guided by the following research questions:

- How can interaction design support awareness and interpretation in horse-human communication during lunging?

- How does real-time technological feedback influence communication, attention and relational awareness between horse and human?

1.2 Ethical considerations

This project involved observations and semi-structured interviews with horse practitioners to better understand how horse-human communication and movement interpretation take place during lunging. All Participants were informed about the purpose of the study before participation and gave informed consent. Participation was voluntary, and participants were informed that they could withdraw at any time without explanation. To protect privacy, all collected material has been anonymized and participants are referred to using pseudonyms. Written consent forms were used during observations and video recordings.

As the project involves horses as part of the studied interaction, ethical considerations also extend beyond the human participants. Although the horses were not direct research participants in the traditional sense, they are central stakeholder in the interaction being studied. The observations were conducted during normal training situations without introducing interventions that could cause stress, discomfort, or changes to the horse's routines. The prototypes developed in this project aims to support awareness and communication without causing discomfort or disrupting the horse- human relationship.

The project also acknowledges the researcher's positionality. As someone with personal experience of horse handling and lunging, the researcher brings prior embodied knowledge into both observations and interpretations. While this provides contextual understanding of the practice, it may also influence interpretation. To address this, reflections from self-observation were combined with observations and interviews with other practitioners to broaden perspective and reduce individual bias.

1.3 State-of-the-art

The understanding of equine movement has traditionally relied on human observation, where trainers, veterinarians and horse owners assess rhythm, symmetry, posture and responsiveness to understand both physical condition and wellbeing. Research shows that lunging commonly used in veterinary practice to observe movement symmetry and identify lameness, as it allows the horse's movement to be studied under circular motion and changing surfaces (Pfau et al., 2014). This highlight lunging as an important site for movement interpretation, but also a demanding one, where subtle changes can be difficult to detect in real time.

At the same time, equitation science has increasingly introduced technological systems such as wearable sensors, motion tracking and physiological monitoring to support more objective assessments of horse's movement and welfare. Randle et al. (2017) discuss how technology

can improve precision and reduce human bias, while also warning that technology should not replace situated understanding and practical knowledge. Similarly, Aydin and Seyidoglu (2023) describe how emerging technologies can support the horse-human relationship by making physiological and behavioural signals more visible, strengthening awareness of welfare and communication.

Within Animal-Computer Interaction, Animals are approached as active participants rather than passive objects of measurement. Mancini (2011) argues for designing technologies with animals as users and stakeholder, while later work on animal-centred ethics emphasizes welfare and consent as central design principles (Mancini, 2016). This perspective shifts technology from being purely diagnostic toward becoming relational and situated.

This project is also informed by more-than-human design perspectives, which challenge human-centred views of interaction by emphasizing interdependence and relationality. Haraway (2015) describes multispecies relationships as processes of "making kin", where beings co-create worlds together. Similarly, Núñez-Pacheco and Rosén (2024) describe noticing as a way of attending to subtle felt experiences and relations, offering a relevant methodological lens for understanding horse-human interaction.

While existing equine technology focus largely on performance, diagnostics and measurement, fewer systems explore how technology can support the lived and embodied communication between horse and human in real time. This project addresses that gap by exploring how interactive systems can support awareness, interpretation, and mutual understanding during lunging, with equine welfare and relational communication at the centre.

1.4 Contributions of the Project

This project aims to contribute to the practice of horse training and welfare by exploring how interactive systems can support awareness and interpretation in horse-human communication during lunging. By focusing on subtle and often overlooked signals such as movement, rhythm, breathing, posture and tension, the project investigates how technology might act as a supportive layer for noticing and reflection, rather than as a tool for control or replacement of human expertise. This may contribute practical insights for riders, trainers and horse owners in how they understand and respond to their horse.

Within the field of interaction design, the project contributes by exploring embedded and physical interaction systems as tools for supporting situated perception in multisensory and embodied practices. Rather than designing for efficiency or optimization, the project investigates how technology can mediate awareness in real-time interaction, raising questions about how feedback should be designed to support attention without disrupting an existing relational dynamic.

The project also contributes to broader discussions in Animal-Computer Interaction and Multispecies Design, by examining how technology can be designed for communication across species. By positioning the horse as an active stakeholder in the interaction, the project contributes knowledge about how interactive systems can support mutual understanding, care and relational awareness in more-than-human contexts.

Finally, the project aims to contribute a critical comparison between horse-human interaction with and without technical mediation. By evaluating whether technology strengthens, changes, or potentially disrupts communication and awareness, the project seeks to contribute a more nuanced understanding of the role of technology in multispecies interaction.

2. Background

2.1 Animal-Computer Interaction

Animal-Computer Interaction (ACI) is an interaction design field concerned with the relationship between animals and technology and with designing interactive systems that consider animals as active participants rather than passive objects. Mancini (2011) describes ACI as a shift away from purely human-centred design, arguing that animals should be understood as users and stakeholders in interaction systems. This perspective is especially relevant in contexts where technology mediate relationships between human and animals, such as training, welfare, rehabilitation and communication.

A central contribution of ACI is the idea that technology should be designed not only for human goals, but also with consideration for the animal's experience, needs and wellbeing. This means acknowledging that animals interact with and experience systems differently than humans, and that their sensory worlds, behaviour and agency must be considered in the design process (Mancini, 2011). In the context of horse-human interaction, this is important because horses are not passive receivers of human actions, but active communicators whose responses shape the interaction.

Ethics is also a central concern within ACI. Mancini (2017) argues for an animal-centred ethics, where animal welfare is not treated as secondary to human goals, but as an essential part of the design process. This includes reflecting on consent, comfort, stress and unintended consequences when introducing technology into animal-related contexts. In this project this perspective is important as the horse is treated as a stakeholder in the interaction, and any technological intervention must aim to support awareness and communication without causing discomfort or disrupting the horse-human relationship.

Wearability is another relevant area within ACI, particularly when designing technology that is physically attached to animals. Paci, Mancini and Price (2020) propose an animal-centred framework for wearability, emphasizing comfort, fit, movement and species-specific bodily conditions. This is relevant for this project as it explores embedded and physical interaction systems that may be integrated into existing horse equipment. Designing for horses means considering not only what data can be captured, but how the technology physically and sensorially affects the horse.

In this project, ACI provides a framework for understanding technology not as a neutral measuring tool, but as an active mediator within horse-human communication. Rather than focusing solely on movement data or performance optimization, the ACI perspective supports a relational and welfare-oriented approach, where the aim is to support mutual understanding between species.

2.2 More-Than-Human and Multispecies Design

Multispecies Design and More-Than-Human Design expand interaction design beyond human-centred perspectives by recognizing that humans exist in ongoing relationships with non-human beings, environments and systems. Rather than treating non-humans as passive resources or background actors, these approaches frame design as something that takes place within interconnected ecologies of living beings. This shift challenges traditional human-centred design by asking how design can support coexistence, care and mutual responsiveness across species (Wakkary et al., 2015).

A key idea within more-than-human design is relationality, where meaning and understanding emerge through relationships rather than existing independency within one individual. Haraway (2015) describes this as "making kin", emphasizing how humans and non-humans co-create worlds together through everyday practices of living, responding and caring. In the context of horse-human interaction, this perspective is relevant because understanding is not simply based on observing the horse from outside, but develops through ongoing interaction, shared routines and mutual adjustment.

Another relevant concept for this project is noticing. Núñez-Pacheco and Rosén (2024) describe noticing as a design approach focused on attending to subtle experiences, relations and felt sensations that are often overlooked in everyday interaction. Rather than prioritizing efficacy or clear information transfer, noticing emphasizes sensitivity and attentiveness. This is particularly relevant in horse-human communication, where small shifts in movement, rhythm, breathing and posture can carry important meaning. In this project, noticing is used both as a methodological lens for observations and as a design principle for exploring how technology might support awareness of subtle signals.

Kimmerer (2013) similarly argues for a relational way of understanding the world, where living beings are approached as active subjects, living beings, rather than inanimate objects. This perspective challenges instrumental relationships with animal and instead emphasizes reciprocity and respect. Applied to this project, this perspective supports moving away from designing for control and toward designing for mutual understanding and care.

In this project More-than-Human and Multispecies Design provide a theoretical framework for understanding lunging as a relational practice rather than simply a training exercise. Through this lens, horse-human interaction can be understood as an ongoing process of communication, interpretation and adjustment, where technology has the potential to support awareness and strengthen relational understanding, rather than replacing or dominating it.

2.3 Equitation science and horse-human communication

Equitation science in an interdisciplinary field that studies horse training and horse-human interaction through learning theory, biomechanics and animal behaviour. The field aims to improve communication, welfare and safety by understanding how horses perceive and respond to human signals (McGreevy & McLean, 2010). Unlike traditional training

approaches that often rely on tacit knowledge and experience, equitation science provides a framework for understanding horse-human interaction in a more systemic way.

A central concept in equitation science is that communication between horse and human is based on signals and responses, where timing and consistency are essential. Horses learn through reinforcement and pressure-release, meaning that the human's body language, positioning, voice and physical cues directly shape the horse's understanding of what is being asked (Mcgreevy & McLean, 2010). This makes horse-human interaction highly dependent on the human's ability to both give clear signals and interpret the horse's responses.

During lunging, this communication becomes especially visible, as the handler works from the ground and relies on voice, body position and the lunge line to guide movement. At the same time, the horse communicates through posture, rhythm, breathing, ear position, tension and willingness to respond. Aydin and Seyidoglu (2023) describe horse-human interaction as a process of interpreting silent communication, where behavioural and physiological signals provide important information about the horse's state and welfare. This perspective aligns with the observations in this project, where participants described continuously scanning the horse's movement and behaviour while adjusting their own actions in response.

Equitation science also highlights that movement quality is closely linked to welfare and physical condition. Irregularities in rhythm, asymmetry, tension or reluctance to move forward may indicate discomfort, imbalance or injury (McGreevy & McLean, 2010). This makes the interpretation of movement an important part of everyday horse care, not only for performance but also for wellbeing.

In this project, equitation science provides a practical and theoretical foundation for understanding lunging as communicative and relational practice. It helps frame horse-human interaction as a continuous exchange of signals, where perception, timing and responsiveness are central to both communication and care.

2.4 Equine Movement assessment and Technology

Technological systems for equine movement assessment have developed significantly on recent years, offering new ways of measuring gait, asymmetry and movement irregularities. Wearable sensor systems, inertial measurement units (IMUs), motion capture, and gait analysis technologies are increasingly used in veterinary practice, rehabilitation and performance training to support objective assessment of movement (Serra Bragança et al., 2018). These technologies provide quantitative data that can complement human observation, especially in identifying subtle asymmetries that may be difficult to detect with the naked eye.

Research has shown that subjective human assessment of equine movement can vary considerably, even among experienced professionals. Keegan et al. (2010) found that evaluations of lameness and movement irregularities are often inconsistent, highlighting the limitations of relying solely on visual assessment. This has strengthened the role of technological systems in equine healthcare, where objective measurements can improve precision and reduce observational bias.

Lunging has become an important context for movement analysis, as circular movement can reveal asymmetries that may not be visible during straight-line assessment. Pfau et al. (2014) show that speed and circle radius influence movement symmetry, making lunging a valuable but complex situation for movement interpretation. This highlights both the opportunities and challenges of assessing equine movement in dynamic, real-world settings.

At the same time, technology within equitation science has expanded beyond clinical diagnostics and into training and welfare contexts. Randle et al. (2017) discuss how technological systems can support understanding of horse movement, physiological responses and behaviour, while emphasizing that technology should be seen as a complement to practical knowledge rather than a replacement for human expertise.

Despite these developments, much of the existing technology focuses on performance optimization, diagnostics and data analysis after the horse-human interaction. Less attention has been given to how technology might support awareness and interpretation during the interaction itself, in real time and within the ongoing communication between horse and human. In the context of lunging, where attention is distributed across movement, rhythm, tension and responsiveness, there is an opportunity for interaction design to explore how technological feedback can support noticing and relational understanding rather than simply producing measurements.

In this project, current equine sensing technologies provide both technical inspiration and a point of contrast. While they demonstrate what can be measured, this project focuses on how measured or sensed information can be translated into embodied and situated feedback that supports communication and care.

3. Methodology

3.1 Design process

This project follows a research through-through-design (RtD) approach, where knowledge is developed through the process of designing, prototyping and reflecting on interactive systems. Research-through-design is an established approach within interaction design, where the act of making is used as a way of investigating complex situations and generation knowledge through design practice rather than only through observation or analysis (Zimmerman et al., 2010). RtD is relevant in this project because the aim is not only to study horse-human communication during lunging, but also to explore how technology might support that interaction.

The overall design process is iterative and exploratory, moving between understanding the situation, generating ideas and testing possible solutions. Boeijen (2013) describes design as a cyclical process where insights continuously shape new design directions, rather than following a fixed linear path. This approach is suitable for this project because the design problem is still emerging through fieldwork and exploration.

The process was structured into four main phases: exploration, ideation, prototyping, and evaluation. These phases are inspired by common interaction design processes, where understanding users and contexts, generating alternatives and evaluating solutions are seen as interconnected activities rather than separate steps (Rogers et al., 2019). Each phase contributes to both the development of the design concept and the knowledge produced in the project.

The exploration phase focused on understanding the context of lunging through apprentice-style observations, interviews, self-observations and video analysis. Semi-structured interviews were used to gain deeper insights into participants experience and interpretation while allowing flexibility around central themes (Brinkmann & Kvale, 2018). Apprentice-style observations were inspired by contextual inquiry, where the participant takes the role of expert and the researcher adapts the role of learner to access situated and practical knowledge through participation and dialogue (Holtzblatt & Beyer, 2016). Video recordings later supported repeated examination to subtle interaction details that may be difficult to fully perceive during live observation (Heath et al., 2010). This phase aimed to uncover how horse-human communication takes place, what handlers pay attention to and what challenges exist in interpreting movement and behaviour.

The ideation phase builds on these findings through divergent sketching and concept generation. Boeijn (2013) emphasizes the importance of exploring multiple alternatives before committing to a solution, which is central to this phase. The goal is to investigate different ways technology could support awareness, communication and movement interpretation.

The prototyping phase focused on materialising selected concepts into prototypes that gradually evolved throughout the design process. Rather than viewing prototypes primarily as low- or high-fidelity representations, Houde and Hill (1997) argue that prototypes can be understood through what aspects of design they investigate: the role of a system, its look and feel or its implementation. Looking back at the process, different prototypes in this project can be seen as exploring these dimensions to varying degrees. Early concept sketches explored possible roles of technology could play in horse-human communication, while Wizard-of-Oz testing and haptic explorations focused more on the experience and interpretation of feedback during interaction. Later iterations increasingly addressed questions related to technical functionality and implementation. The prototypes therefore functioned as research artefacts that supported exploration and reflection throughout the design process.

Finally, the evaluation phase involved testing the prototypes in the context of lunging, comparing horse-human interaction with and without technological support. This phase focus on understanding how the technology influences perception, communication and awareness in practice.

This process was iterative with insights and reflections continuously moving between exploration, synthesis, ideation, prototyping and evaluation. Figure 1 illustrates the iterative nature of the design process. Rather than progressing through isolated stages, observation, reflections and prototype testing continuously generated new insights that formed subsequent design decisions and occasionally led back to earlier phases

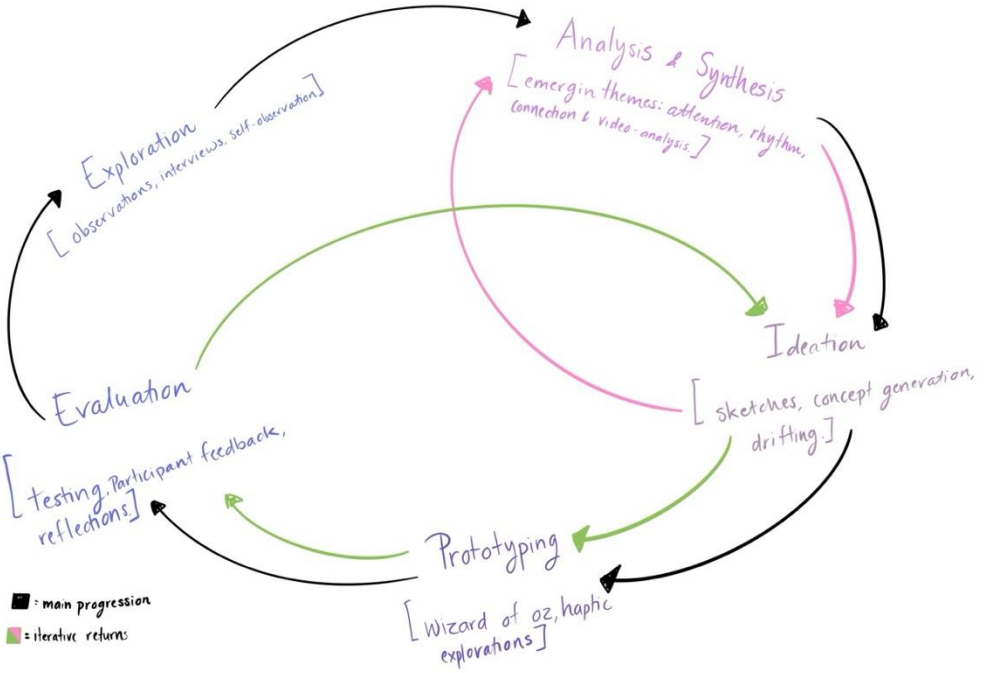


Figure 1. Iterative RtD process used throughout the project. While the process folowwed an overall progression, findings and reflections continuously informed design decisions and led back to earlier phases.

3.2 Research Methods

This project uses a qualitative research approach to explore horse-human communication and movement interpretation during lunging. Since the project focuses on perception, embodied communication and situated interaction, methods were chosen to capture both spoken reflections and physical practices in context.

Since this project explores embodied communication and situated interaction, multiple qualitative methods were combined to capture both participants experiences and the practices occurring during lunging sessions. Using several methods also supported the examination of the interaction from different perspectives, allowing spoken reflections to be considered alongside observations of bodily communication and behaviour.

Field observations were used to study lunging sessions in their natural setting. Observation is useful in interaction design research because it allows practices and behaviours to be studied as they occur, rather than being reconstructed afterward (Rogers et al., 2019). In this project, observation focused on how handlers attend to movement, communicate with the horse and interpret behavioural and bodily signals.

Apprentice-style observation was used to access tacit knowledge from experienced horse practitioners. Inspired by contextual inquiry, this approach positions the participant as the expert and the researcher as a learner, allowing practical and embodied knowledge to emerge

through situated action and explanation. Holtzblatt and Beyer (2016) describe this as a master-apprentice relationship, where the researcher learns by observing, asking questions and following the participant's practice in context. In this project, participants were asked to explain their actions and decisions while lunging, making implicit knowledge visible through verbalization during practice.

Semi-structured interviews were conducted alongside observation with four participants to gain deeper insights into participants experiences, interpretations and thought a technology in horse practice. Semi-structured interviews allow flexibility while still focusing on key themes, making them suitable for exploring individual perspectives (Rogers et al., 2019). This method of interviewing were chosen as they allow flexibility while maintaining focus on central themes and participants experiences (Brinkmann & Kvale, 2028).

Self-observation was also used a part of the research process, where the researcher participated directly in lunging sessions while reflecting on personal embodied experiences of the interaction. As the researcher has prior experience with horse handling and lunging practices, this approach provided access to situated and practice-based knowledge that supported interpretation of subtle aspects of the interaction. The method offered firsthand insights into physical and sensory dimensions of communication, including rhythm, body positioning and attentional shifts.

3.3 Analysis Methods

The collected material was analysed through a qualitative and iterative process inspired by thematic analysis (Braun & Clarke, 2006). Field notes, interview material, self-observations and video recordings were reviewed multiple times to identify recurring themes, patterns and tensions related to horse-human communication during lunging.

Video recordings functioned as complimentary analytical tool throughout the process. Revisiting recordings repeatedly made it possible to examine interaction sequences and subtle bodily details that where difficult to fully perceive during live observation. Kendon (1990) describes interaction as an organised bodily process where meaning emerges through movement, orientation and visible action in relation to others. Similarly, repeated viewing supported closer examination of movement patterns, behavioural responses and shifts in interaction between horse and handler.

Particular attention was given to how handlers perceived and interpreted subtle communicative signals, including rope tension, rhythm, breathing, posture, ear orientation and responsiveness. Attention was also directed toward description of relaxation, tension, attentiveness and willingness to engage. Observations and interview reflections were compared across participants and considered alongside insights from self-observation sessions.

The analytical process remained closely connected to the ongoing design process. Emerging themes and tensions were continuously translated into design directions and interaction qualities explored through sketching and prototyping. This includes themes such as rhythm,

instability, embodied awareness, attentiveness and the balance between support and control. Following Kolko (2010), synthesis was used to connect insights from fieldwork, theory and design exploration into meaningful design concepts.

3.3 Ideation and Sketching Methods

The ideation phase focused on exploring different ways technology could support connection in horse-human interaction during lunging. Rather than aiming for a single solution immediately, the process involved generating and comparing multiple concepts through sketching, diagramming, low-fidelity prototyping and reflective iteration on emerging findings. Ideas were repeatedly revisited and modified based on observations, participant feedback and prototype testing. For example, early concepts involving screen-based interaction were reconsidered after observations suggested that visual attention could disrupt the sense of connection with the horse. Similarly, haptic interaction patterns were continuously adjusted throughout the prototyping process in response to participants interpretation and experiences of feedback. Inspired by research-through-design approaches, sketches functioned both as design proposals and as a way of thinking through the relational and embodied qualities of the interaction.

Early ideation explored multiple forms of feedback, including visual, auditory and haptic interaction. These concepts were developed in response to findings from fieldwork and self-observation, where attention, rhythm, bodily awareness and subtle communication emerged as central aspects of the interaction. These sketches were used to explore how technology might support awareness without disrupting the existing relationship between horse and handler.

The ideation process was iterative and non-linear. Rather than following a fixed path, the project moved between observations, sketches, prototype tests and reflection. This process was mainly inspired by what Krogh, Markussen and Bang (2015) describe as serial and probing forms of drifting. Serial drifting was relevant because one design experiment led to the next, where insights from observations and prototype testing gradually redirected the concept. Probing drifting was also relevant because the project explored an uncertain and emerging design space, where prototypes were used to investigate what kinds of interaction might be meaningful rather than to test a predefined solution. Initial ideas focused more directly on movement tracking and performance-related feedback but gradually shifted toward relational and embodied interaction. Through this process, haptic feedback emerged as a promising direction because it allowed the interaction to remain bodily and peripheral, without requiring the handler to shift attention towards screens or visual interfaces.

To explore the experiential qualities of haptic interaction before developing a fully functioning sensing system, Wizard-of-Oz prototyping was used. Wizard-of-Oz is a prototyping method where participants interact with a system that appears autonomous, while parts of the interaction are manually controlled behind the scenes (Sharp et al., 2019). In this project, haptic patterns and interaction states were manually adjusted and tested during lunging sessions. The purpose was not to evaluate technical precision, but to investigate how different haptic rhythms, intensities and patterns were experienced in relation to connection, attention and communication during lunging.

Sketches, interaction diagrams and prototype iterations were continuously documented throughout the process and later used as material for reflection and synthesis.

3.4 Prototyping Methods

The prototyping phase focused on exploring how embodied and haptic interaction could support awareness and connection during lunging. Following research-through-design approaches, prototypes were treated as exploratory tools for investigating experimental qualities of interaction rather than only functional solutions (Zimmerman et al., 2007). The prototypes were developed iteratively through sketching, physical computing and repeated testing throughout the design process. Following Boeijen et al. (2013), sketches and prototypes functioned as external representations that supported exploration and reflection during development. The process also reflected serial drifting as described by Krogh, Markussen and Bang (2015), where insights from one prototype iteration informed subsequent developments. Changes in vibration rhythms, interaction states and feedback behaviour gradually redirected later prototype iterations as understanding of connection and attentiveness evolved.

The prototypes were developed through iterative low fidelity experimentation using physical computing components, including Arduino Nano 33 Sense boards, vibration motors and manually controlled inputs. Early prototypes explored different forms of feedback, including variations in vibration intensity, rhythm, regularity and absence of feedback. The purpose was to investigate how different haptic patterns could be experienced and interpreted in relation to the ongoing horse-human interaction.

Wizard-of-Oz prototyping was used to simulate interaction states before implementing automated sensing (Sharp et al., 2019). In these prototypes, haptic feedback was manually controlled through a potentiometer connected to a separate microcontroller. This allowed different relational states, such as unstable connection, steady connection or absence of connection, to be explored without relying on technically accurate sensing systems. The method made it possible to focus on the experimental and relational qualities of the interaction rather than technical precision.

The prototypes were continuously adjusted based on observations and testing throughout the design process. Changes in pulse timing, vibration intensity and rhythmic patterns were explored to investigate how haptic feedback could remain subtle and non-disruptive within the ongoing interaction. In this project, peripheral feedback refers to feedback that remains within the handler's background awareness rather than becoming the primary focus of attention. The intention was that the handler's attention should remain directed towards the horse and the interaction itself, while the haptic feedback functioned as a supportive layer of information. This relates to Dourish's (2001) perspectives on embodied interaction, where meaning emerges through situated bodily engagement rather than through detached interpretation of information. Instead of redirecting attention towards screens or explicit instructions, the feedback was therefore designed to become integrated into ongoing bodily interaction between horse and handler.

3.5 Evaluation Methods

The prototypes were evaluated through situated testing, reflective observation and informal participant feedback during prototype interactions. Since the project focused on horse-human communication and interaction experiences during lunging, the evaluation emphasized participant's experiences and interpretations rather than quantitative performance measures.

The evaluation explored how the haptic feedback was experienced in relation to attention, awareness and the previously identified qualities associated with connection between horse and handler during lunging. These qualities included responsiveness, attentiveness, rhythmic consistency, willingness to engage and participant's description of the horse being "with" the handler. Rather than functioning as objective measurements, these qualities served as interpretive indicators that informed reflection on how the prototypes influenced the ongoing interaction. Attention was also given to whether the feedback was experienced as supportive, distracting, calming, intrusive or meaningful, and whether participants could distinguish and interpret different haptic states.

Evaluation took place iteratively throughout the design process rather than only at the end of the project. Adjustments to vibration intensity, rhythm, timing and regularity were made in response to observations and participant experiences during testing sessions. In line with research-through-design approaches, evaluation therefore became part of the ongoing development process rather than a separate activity conducted after implementation (Zimmerman et al., 2007).

Because the project explores interaction within a multispecies context, evaluation also included reflection on equine welfare and the potential impact of the technology on the horse-human relationship. Consideration was given to whether the prototypes risked shifting attention away from the horse or introducing stress, interruption or over-reliance on technological feedback. The evaluation therefore focused not only on usability, but also on how the interaction aligned with the project's aim of supporting relational awareness and care.

4. Design Process

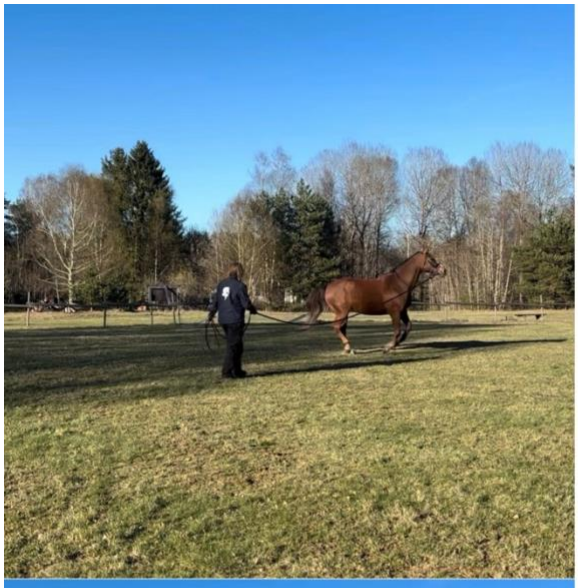
4.1 Fieldwork and Observations

The design process began with field observations, interviews, self-observation and apprentice-style participation during lunging sessions. These activities aimed to develop a situated understanding of how communication, attention and movement are experiences between horse and handler during lunging. Observations were conducted in familiar environments for both horses and participants, including settings where training and handling normally took place. The researcher also had previous experience with horse handling and lunging practices and therefore entered the field with existing embodied and contextual knowledge of the practice. This enabled closer participation and supported interpretation of subtle aspects of the interaction that may have been difficult for an external researcher to access. At the same time,

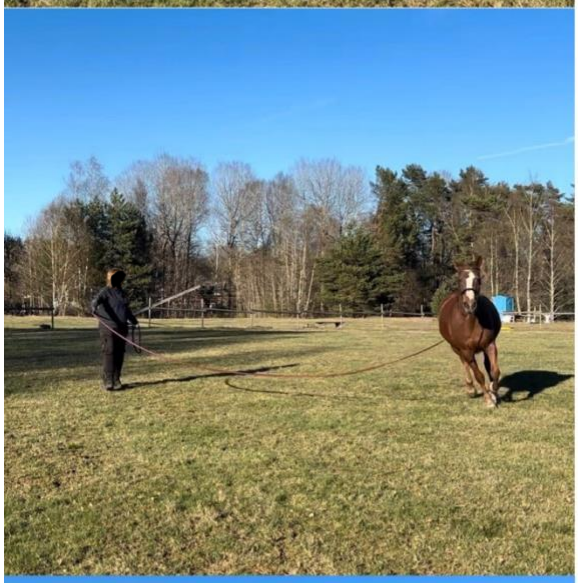
this position required ongoing reflection regarding assumptions and interpretations throughout the process.

Video recordings from the sessions were used as a complimentary analytic tool. Revisiting the recordings repeatedly made it possible to study subtle changes in timing, rhythm, posture, responsiveness and bodily communication that were difficult to fully perceive in real time. The recordings supported reflection on how interaction patterns shifted during the sessions and helped identify moments where attention, rhythm or communication changed between horse and handler.

During the observations, it became clear that lunging involves continuous interpretation of subtle bodily and behavioural signals (see figure2). Handlers described paying attention to aspects such as rope tension, rhythm of movement, breathing, ear orientation, posture, tempo and responsiveness to cues. Rather than relying on a single signal, communication emerged through an ongoing combination of bodily awareness, timing and interpretation. All participants described the importance of the horse feeling “with them”, often referring to moments where the interaction became calmer, more rhythmic, attentive or responsive.



In this frame one can see that the horse is looking elsewhere, out in nature.
The horse appears less attentive toward the handler and more focused on the surrounding environment
The head is carried high, indicating some form of tension.
She's not fully distracted, she still holds the gate the handler has asked for.



In this frame the horse is more focused on the handler.
She has her ear pinned on the handler and is listening in on the cues the handler is giving.
The horse is starting to move more rhythmic but still carries her head a bit high, indicating that she is not relaxed yet.



In this frame the horse is focused on the handler, her ear is pinned on the handler.
The horse carries herself in a different composure now. Head a bit lower and moving more through her body. Potentially indicating a more relaxed and engaged state.
You can almost see how the handler also becomes more relaxed.

Figure 2. Frames from video analysis illustrating changes in percieved attentiveness and connection during lunging. The images show how interpretation emerged through combinations of bodily cues, rather than from isolated signals alone. See full video here: https://youtu.be/zepmcO34X5o

The apprentice-style observations also highlighted how attentiveness was expressed bodily through posture, orientation and movement within the interaction. The handler continuously adjusted their position and bodily orientation in relation to the horse, often remaining visually and physically oriented towards the horse throughout the interaction. Video analysis further revealed how the horse's attention could shift between the environment, the handler and surrounding activity while remaining partially engaged in the interaction (See Figure 3 and Appendix B). These observations reinforced the understanding of lunging as a dynamic and relational process shaped through ongoing bodily attentiveness rather than fixed or isolated signals. This relates to Dourish's (2001) perspective on embodied interaction, where meaning emerges through situated bodily engagement and interaction with the surrounding environment rather than through detached interpretation alone.

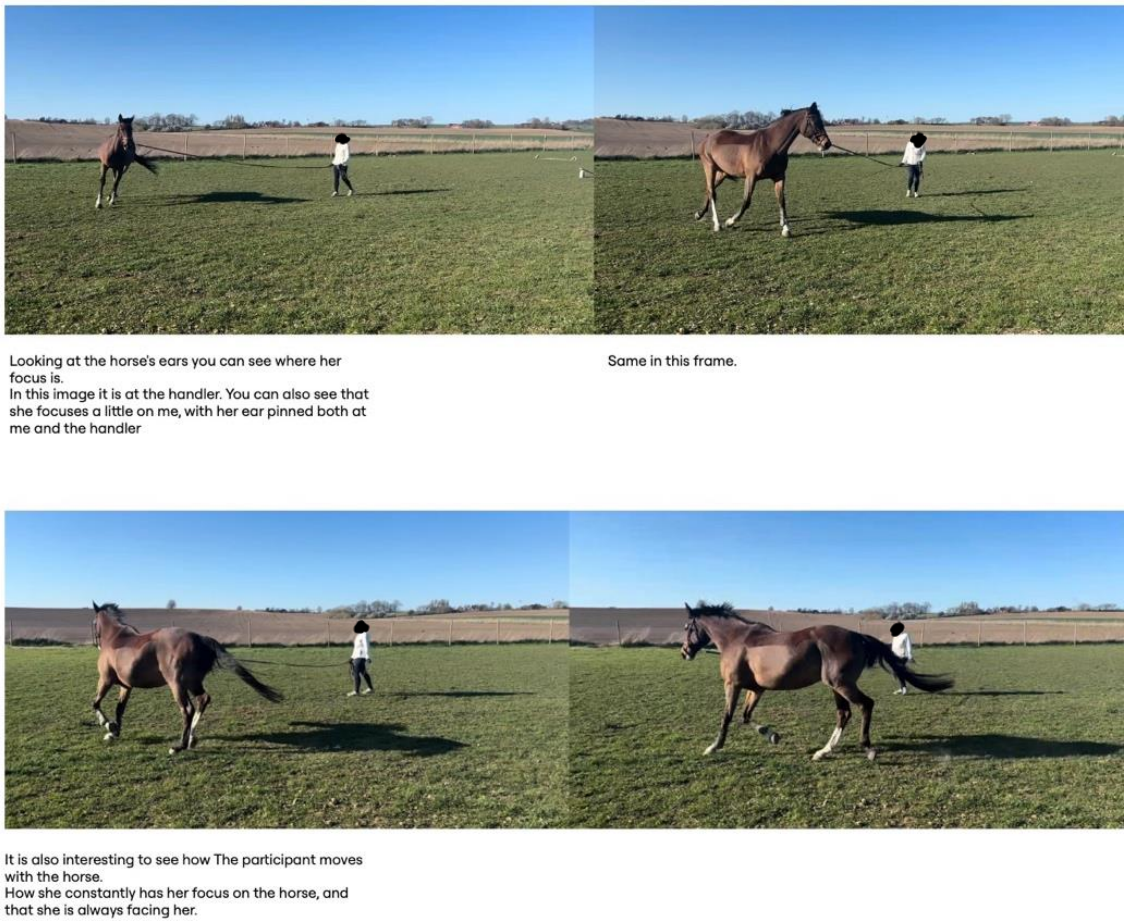


Figure 3. Frames from apprentice-style observation and video analysis during a lunging session. This part of the sequence illustrates how attentiveness and communication were expressed through bodily orientation, posture and movement. See Appendix B for full analysis and watch the video here: https://youtu.be/teIXvGLozp4

The observations also highlighted how demanding this attentiveness can be. Handlers simultaneously monitored the horse's movement, emotional state, surroundings and their own bodily communication. Through self-observation sessions, it became apparent that shifting attention towards a phone or screen disrupted the sense of awareness and connection within the interaction. This became an important insight for the design process and contributed to the decision to avoid screen-based interaction in later concepts and prototypes.

Another recurring theme was that communication during lunging was often described relationally rather than technically. Participants rarely spoke about precision alone, but instead described trust, attentiveness, willingness to engage and mutual understanding as important qualities in the interaction. At the same time, participants also described uncertainty in how to interpret subtle signals from the horse, particularly in moments where tension, distraction or loss of focus emerged gradually rather than clearly.

These observations became central to the later synthesis and ideation phases. In particular, the importance of rhythm, attentiveness, embodied awareness and non-disruptive interaction influenced the development of the haptic prototype concepts explored later in the project.

4.2 Analysis and Emerging Themes

Following the field observations, interviews, self-observation sessions and video-analysis, the collected material was reviewed and synthesised to identify recurring themes and tensions within the horse-human interaction during lunging. Rather than focusing isolated behaviours or measurable performance outcomes, the analysis explored how communication and connection were experienced and interpreted by handlers in practice.

One of the strongest recurring themes was the importance of attentiveness and ongoing awareness during lunging. Participants described the interaction a highly dependent on continuously noticing subtle shift in the horse's movement, rhythm, posture, focus and responsiveness. Communication was often described as fragile and dynamic, where small changes in attention or bodily communication could influence the interaction significantly.

The apprentice-style observations also revealed differences in how participants interpreted and prioritised bodily signals during lunging. Although all participants observed similar interaction contexts, they often focused on different aspects of movement and communication. Some participants paid greater attention to posture, shoulders and rhythm, while the other focused more on pelvic movement, attentiveness or positioning. This suggested that interpretation during lunging is highly situated and influenced by individual experience, attentional focus and embodied understanding rather than fixed or universally agreed-upon indicators. Table 1 presents a comparison of aspects participants noticed across multiple apprentice-style observation sessions.

What was noticed	Lunging session 1	Lunging session 2	Lunging session 3
Position of Tail	Yes	No	No
Position of Head	Yes	Yes	Yes
Position of Legs	Yes	Yes	Yes
Movement of ears	Yes	Yes	Yes
Breathing	Yes	Yes	No
Engaged core	Yes	No	No
Where person is positioned/Own body language	No	Yes	Yes
Relaxation	Yes	Yes	Yes
Hind activation	Yes	Yes	Yes
Noticing differences in direction	Yes	Yes	No
Engaged back	Yes	No	No
Overall Muscles	Yes	Yes	Yes
Rytm	Yes	Yes	No
Pelvis	No	Yes	Yes
Shoulders	Yes	No	No
Hooves	No	No	Yes

Table 1

During self-observation sessions, video recordings and annotated screenshots were used to reflect on subtle shifts in posture, rhythm, responsiveness and attention during lunging. The process made it possible to compare moments of perceived connection and disconnection within the interaction. In particular, moments where the researcher shifted attention toward recording or external tasks highlighted how quickly the sense of connection and attentiveness could change during the interaction (See figure 4). The annotated screenshots also illustrated how interpretations of connection emerged through combinations of bodily cues, including ear orientation, posture, movement quality and overall composure rather than isolated signals.

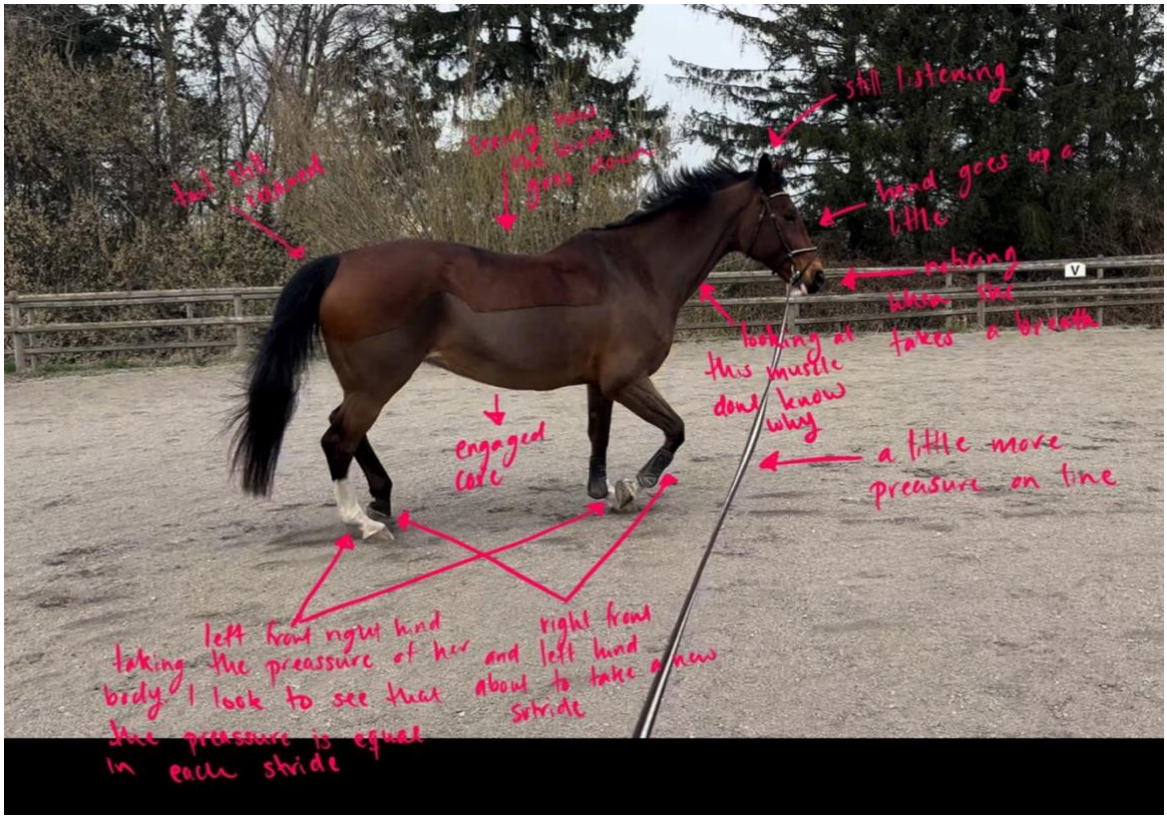


Figure 4. Self-observation and video analysis conducted during lunging sessions. The annotated document reflections on rhythm, posture, attention, responsiveness, and perceived connection between horse and handler across different moments in the interaction. See full analysis in Appendix B & full video here: https://youtu.be/41U4ruA3qnw

Another recurring theme concerned rhythm and consistency. Moments where the horse was described as "being with" the handler were frequently associated with stable rhythm, responsiveness and a calmer quality of movement. In contrast, irregular rhythm, inconsistent tempo, distraction, or delayed response were often interpreted as signs of reduced connection or attentiveness within the interaction. These findings later became important in the development of the haptic interaction patterns explored in the prototypes.

Based on observations, interviews, self-observation sessions and video analysis, several recurring qualities associated with experiences of connection between horse and handler were identified. These qualities were not translated as objective measurements but as interpretive indicators that later informed prototype development and evaluation.

- **Responsiveness** – The horse responds consistently to cues and adjustments from the handler
- **Attentiveness** – Both the horse and handler remain engaged and responsive toward one another throughout the interaction.
- **Rhythmic Consistency** – Movement and tempo remain stable and coordinated.
- **Willingness to Engage** – The horse appears attentive and cooperative rather than distracted or resistant

- **Reduced signs of Tensions or Distraction** – Fewer interruptions, delayed responses or shifts in attention.
- **Perceived Connection** – Participants describe the interaction as calm, synchronized or as the horse being "with them".

These qualities later served as reference points when evaluating whether prototype interactions supported or disrupted the ongoing horse-human relationship.

The analysis also highlighted a tension between support and control in relation to technology. While participants expressed interest in tools that could support awareness and communication, there was also hesitation toward systems that risked becoming overly corrective, performance-focused or distracting. Participants generally described a desire for technology that could support attentiveness without replacing embodied interpretation or shifting focus away from the horse itself.

Together, these themes shifted the project toward relational and embodied perspectives on interaction. Instead of focusing primarily on measurement or optimisation of movement, the findings suggested an opportunity to explore how technology might support attentiveness, awareness and ongoing communication within the horse-human relationship.

4.3 Synthesis and Design Directions

The synthesis process involved bringing together findings from observations, interviews, self-observation and video analysis to identify patterns and tensions that could inform later design decisions. Following Kolko (2010), synthesis was used not only to organize collected material but also to transform observations into meaningful design directions and interaction qualities. Rather than focusing on isolated behaviours or measurable performance outcomes, the synthesis explored how communication and connection were experienced and interpreted during lunging.

Several tensions emerged throughout the analysis and became important during the development of design concepts. These tensions included support and control, attentiveness and distraction, measured and relational awareness, as well as technical feedback and embodied communication, as seen in figure 5. Rather than functioning as fixed categories, these tensions became reflective tools for considering how different technological interventions could shape the interaction between horse and handler.

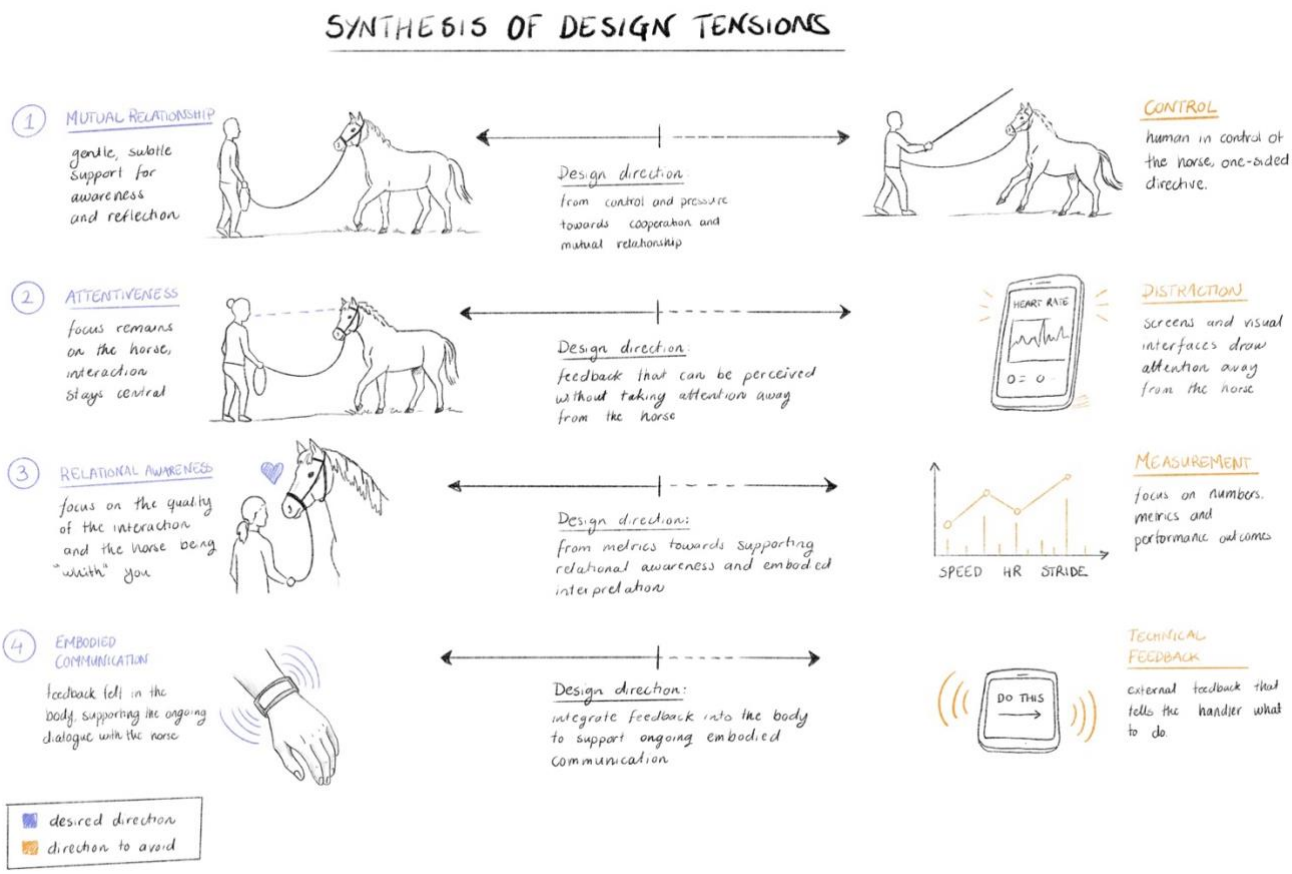


Figure 5. Design tensions explored during synthesis that guided prototype development and interaction design decisions.

One of the clearest insights from the fieldwork was that communication during lunging is already highly embodied and relational. Participants rarely described interaction as relying on isolated cues or technical precision. Instead, they described paying attention to combinations of movement, rhythm, bodily posture, responsiveness and attentiveness that continuously changed throughout the interaction. This suggested that technological support should avoid replacing existing forms of interpretation and instead work alongside them.

Another important synthesis outcome concerned attentional focus. Observations and self-observation sessions indicated that directing attention toward phones or visual interfaces disrupted the ongoing interaction and reduced awareness of subtle changes in the horse's behaviour. This shifted the design direction away from screen-based concepts and toward forms of interaction that could remain integrated into the ongoing activity.

Rhythm also became an important consideration during the synthesis process. Across observations and participant reflections, rhythm was frequently described together with attentiveness, responsiveness and experiences of the horse being "with" the handler. Rather

than being interpreted as an isolated indicator, rhythm appeared as one of several qualities through which participants made sense of the ongoing interaction. This later informed prototype explorations of different haptic patterns, where rhythm was investigated as a way of supporting awareness of changing interaction qualities.

The synthesis process also highlighted ambiguity as an important quality of horse-human communication. Participants sometimes interpreted the same movement patterns differently, and observations suggested that interaction rarely involved clear objective indicators. Rather than treating this ambiguity as a problem to eliminate, it was considered quality to preserve within the design process. This shifted the focus away from systems intended to provide definitive answers and toward technologies that could support reflection and awareness.

Table 2 summarises how recurring findings from the fieldwork and analysis were translated into broader design directions that later guided the ideation and prototyping phases of the project.

Findings from Fieldwork	**Interpretation**	**Design direction**
Handlers relied on subtle bodily signals	Communication is embodied and relational	The design should support bodily awareness
Rhythm and consistency were associated with connection	Interaction quality is experienced temporally	Explore interaction through rhythm and timing
Screens disrupted attentiveness during lunging	Visual interfaces redirect focus away from the horse	Avoid screen-focused interaction
Handlers continuously shifted attention between multiple cues	Interaction already involves high cognitive and bodily attentiveness	Design should remain peripheral and non-demanding
Participants described uncertainty in interpretating the horse	Communication involves ongoing interpretation rather than fixed meaning	Design should support reflection rather than explicit instruction
Participants valued mutual understanding over performance metrics	The interaction was framed relationally rather than technically	Focus on relational awareness rather than optimisation
Lunging was sometimes described as routine-based or inattentive	Attention can shift during repetitive interaction	Explore designs that encourage attentiveness and presence

Table 2

Together, these outcomes shifted the project away from performance-oriented equine technology and toward a more relational design approach. This led to the exploration of haptic and embodied interaction concepts that could remain subtle, peripheral and integrated within the bodily communication already present between horse and handler during lunging.

4.4 Ideation and Concept Exploration

Following the synthesis process, themes identified during fieldwork and analysis were translated into potential design opportunities. Rather than directly developing a single solution, the findings were used as starting points for asking what kinds of interactions could emerge if different qualities of horse-human communication were made noticeable or explored through technology.

Inspired by approaches to divergent design exploration described by Boeijen et al. (2013), the process aimed to generate multiple possibilities before narrowing the design space. Themes from observations including attentiveness, rope tension, rhythm, breathing, responsiveness, environmental distraction and bodily posture became points of departure for later concept development, see Figure 6.

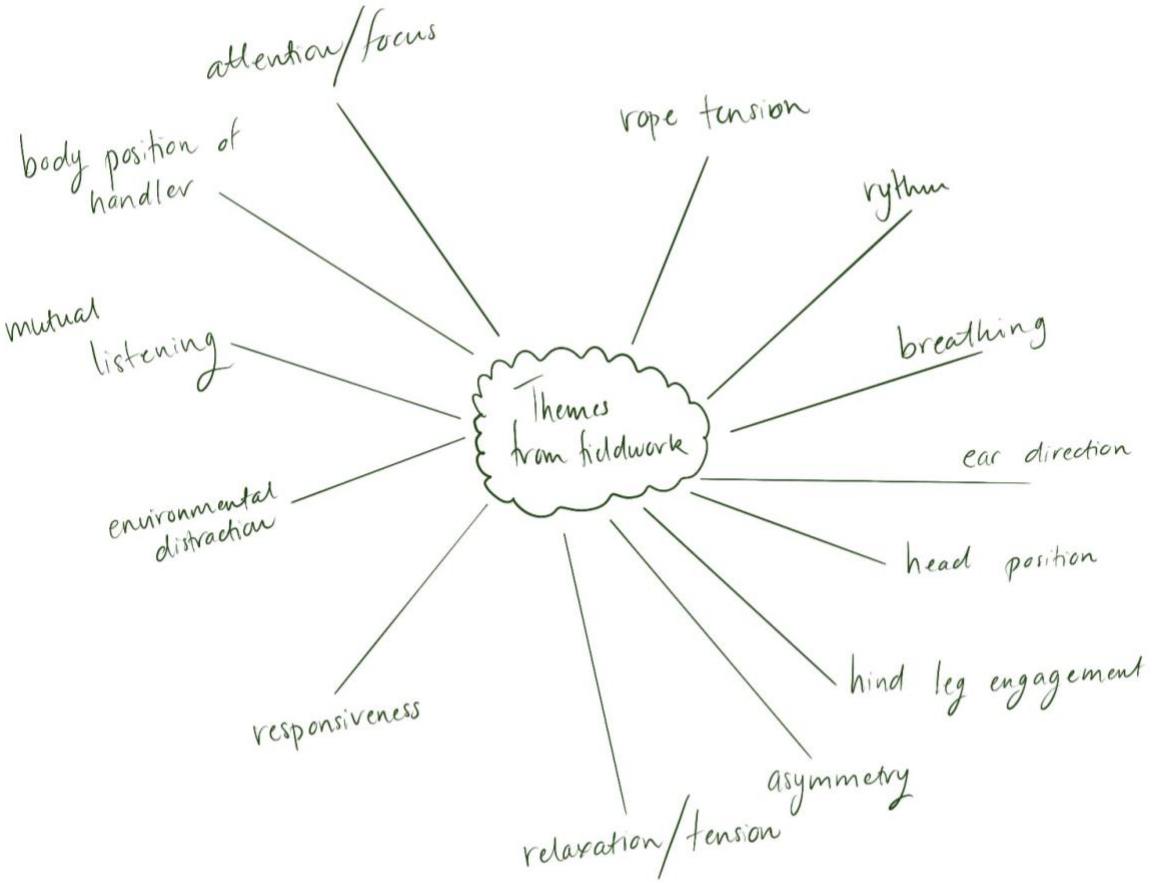


Figure 6. Themes from feildwork translated into potential areas for design exploration.

To avoid prematurely committing to a specific solution, multiple concepts were explored in parallel. Following divergent ideation approaches (Boeijen et al., 2013), concepts addressed different perspectives including communication, movement, welfare, relational interaction and speculative design direction, as seen in Figure 7. Rather than functioning as proposed final systems, these concepts acted as exploratory artefacts used to investigate alternative questions and assumptions.



Figure 7. Early concept exploration made from themes identified during fieldwork.

The concepts varied considerably in their intentions. Some focused on making existing bodily signals more noticeable, while others explored broader questions regarding communication, welfare or technological mediation within horse-human interaction. Several concepts also functioned as speculative explorations questioning assumptions regarding measurement and technological intervention.

During the ideation process, selected concepts were developed further through sketches and interaction scenarios. The sketches functioned as exploratory tools rather than representations of final solutions, allowing design ideas to be reflected upon and compared.

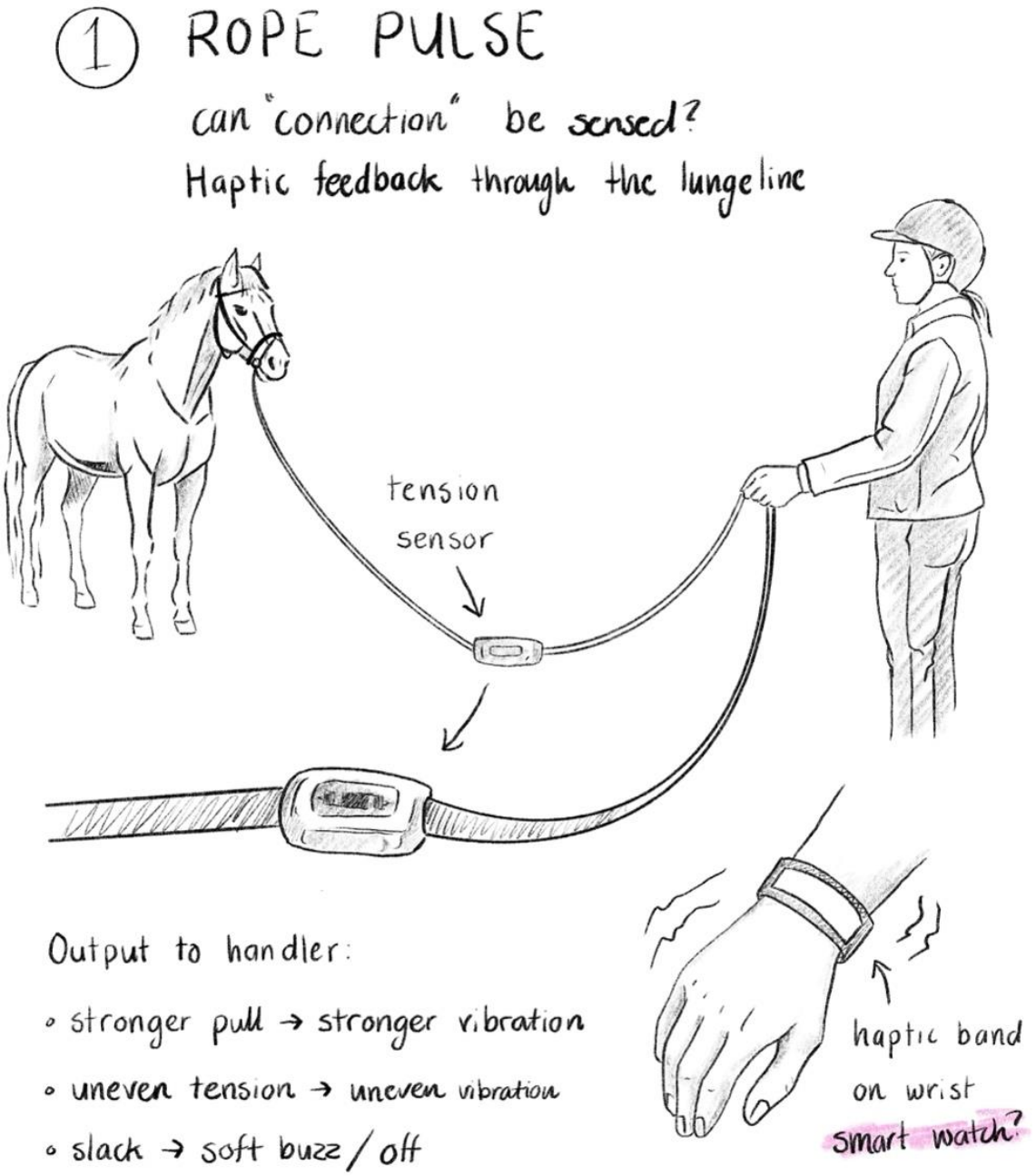


Figure 8. Rope Pulse concept exploring whether changes in rope tension could be translated into haptic feedback.

Rope pulse, as seen in Figure 8, explored whether characteristics of rope interaction could be sensed and translated into haptic feedback. The concept investigated whether tension changes, slack or uneven pulling patterns could function as indicators of changing interaction qualities.

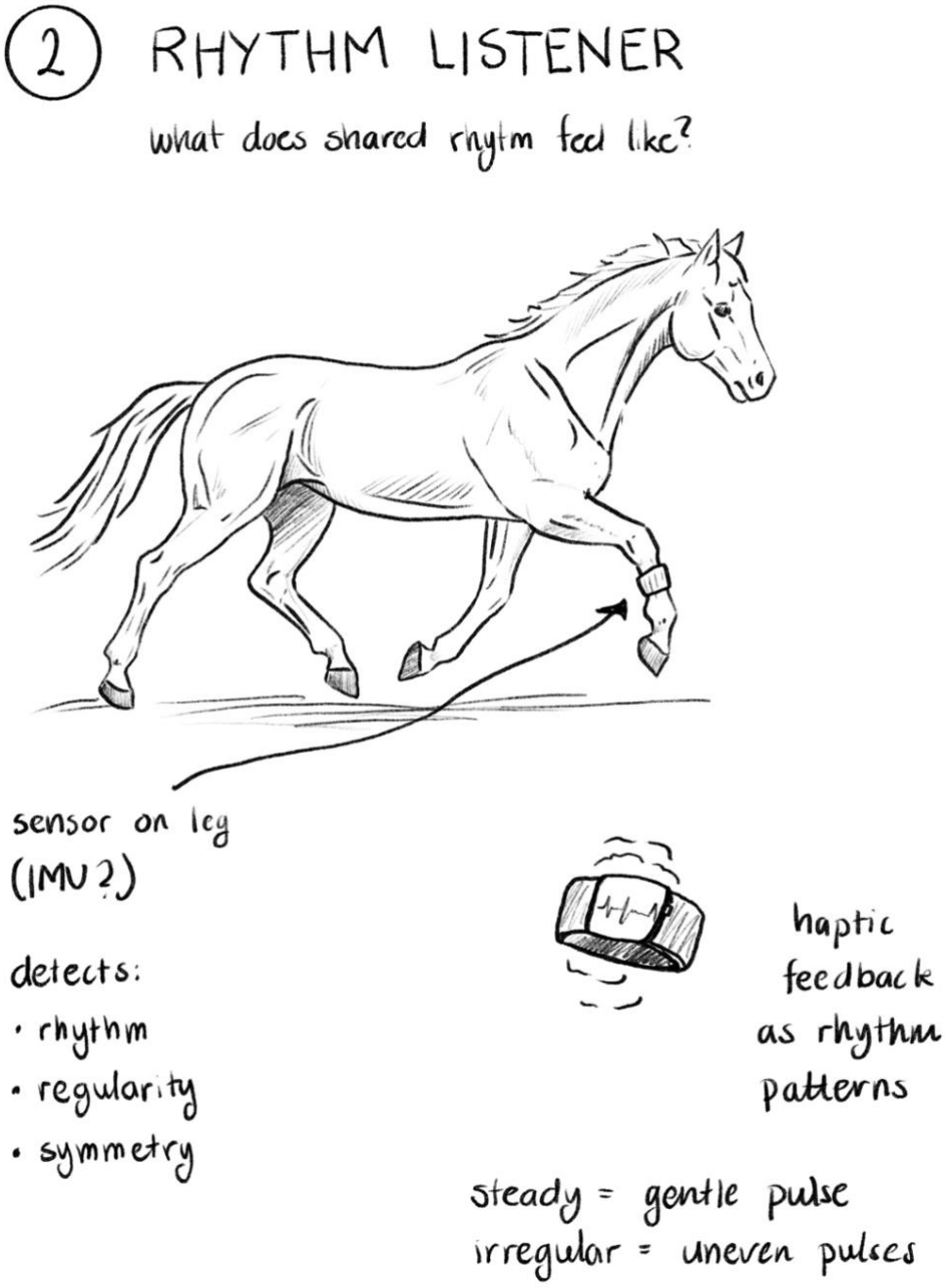


Figure 9. Rhythm Listener concept investigating rhythm as a source of interaction feedback.

Rhythm listener, Figure 9, focused on movement rhythm and regularity. Rather than measuring movement performance, the concept explored whether rhythm could become a way of supporting awareness of interaction qualities and bodily coordination.

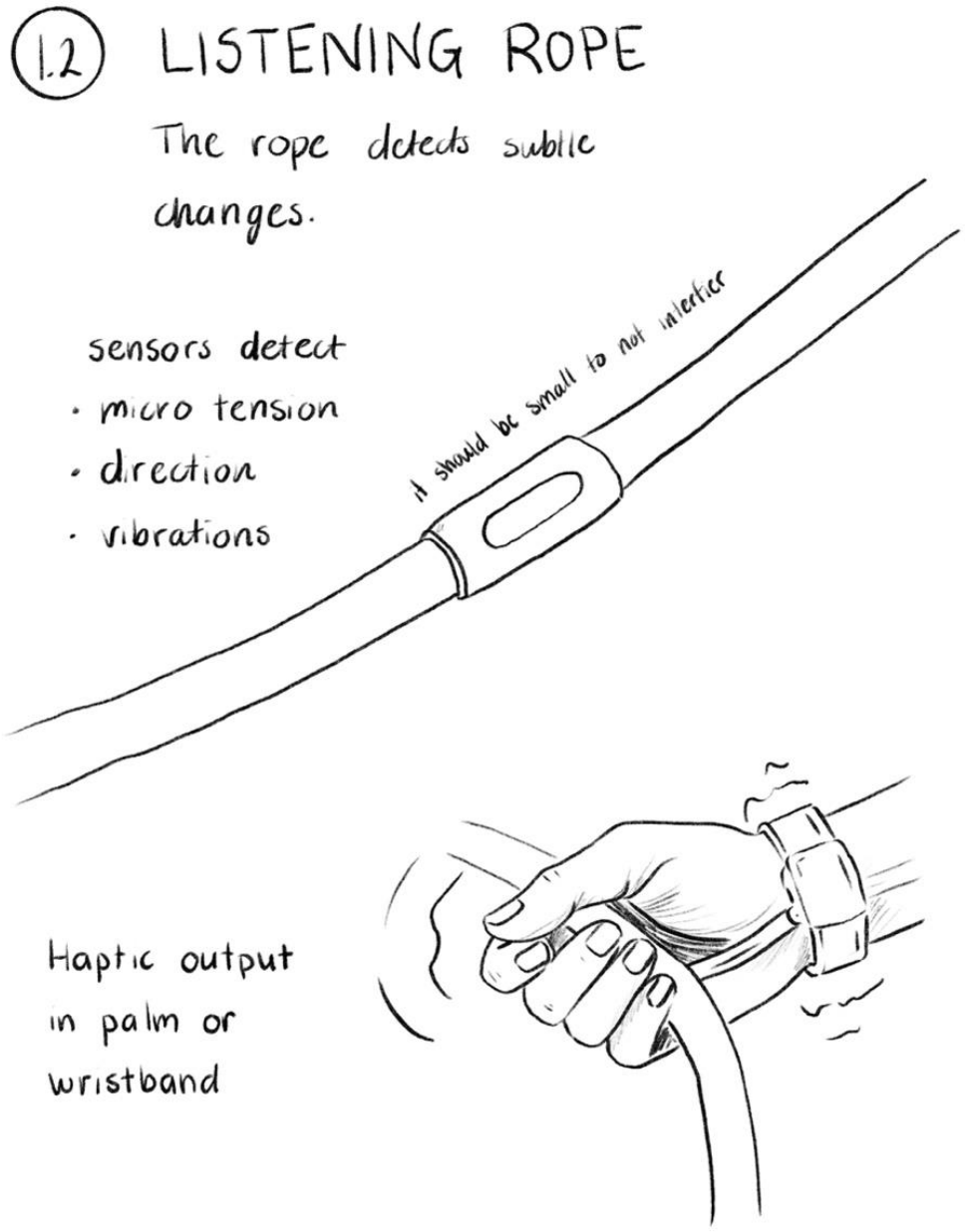


Figure 10. . Listening Rope concept exploring subtle bodily interaction through the lunge line.

Listening rope, Figure 10, explored the possibility of making subtle changes within the interaction more noticeable without introducing additional interfaces or explicit instructions.

Across concept exploration, a recurring concern involved attention and bodily engagement. Earlier observations and self-observation sessions had suggested that handlers relied on continuous attentiveness toward the horse, including subtle shifts in movement, posture, responsiveness and rhythm. Concepts relying on visual displays or explicit information presentation increasingly appeared to risk redirecting attention away from the ongoing

interaction and toward the technology itself. This created a design tension between providing information and maintaining attentional presence within the horse-human interaction.

Inspired by Dourish's (2001) perspective on embodied interaction, the design process gradually shifted toward forms of interaction that could remain integrated into ongoing bodily activity rather than functioning as separate layers of information. Dourish argues than meaning and understanding emerge through situated bodily action, suggesting that interaction should support existing practices rather than interrupt them. Within this project, this perspective became relevant because horse-human communication was observed to already rely on continuously bodily interpretation and responsiveness.

Through this process, haptic feedback emerged not as a predefined goal but because of the design exploration itself. Unlike screen-based concepts, haptic feedback offered the possibility to remaining present in the periphery of attention while allowing the handler to maintain focus on the horse. Rather than communicating objective measurements or corrective instructions, the intention gradually shifted toward supporting awareness, interpretation and reflection within ongoing interaction.

The Rope Pulse concept as seen in figure 8, was selected for further exploration because it aligned more closely with the design requirements identified in table 2. In particular, the concept supported peripheral interaction by allowing feedback to remain integrated within the ongoing activity rather than requiring visual attention or disrupting the interaction. It also offered the possibility of supporting awareness of relational qualities while maintaining the handlers focus on the horse. While other concepts explored relevant aspects of interaction, they either relied more strongly on isolated measurements or risked shifting attention away from the embodied interaction itself. Rope Pulse was therefore considered the most promising direction for further development and prototyping.

4.5 Prototype Development

Based on the concept explorations presented in the previous section, haptic interaction was selected for further investigation through prototyping. Rather than aiming to develop a fully functioning technological system, the prototypes were treated as exploratory artefacts used to investigate how haptic feedback could influence awareness and interaction during lunging. Following research-through-design perspectives, prototypes functioned as tools for inquiry, allowing ideas and interaction qualities to become tangible and testable through making and reflection (Zimmerman et al., 2007).

The purpose of the prototype phase was therefore not to evaluate technical precision or sensing accuracy but rather investigate how different forms of haptic feedback could be experienced in relation to connection, attentiveness and horse-human communication. The prototypes functioned as a way of exploring questions that had emerged from earlier fieldwork and synthesis, including how technology could support awareness while remaining integrated into the ongoing interaction.

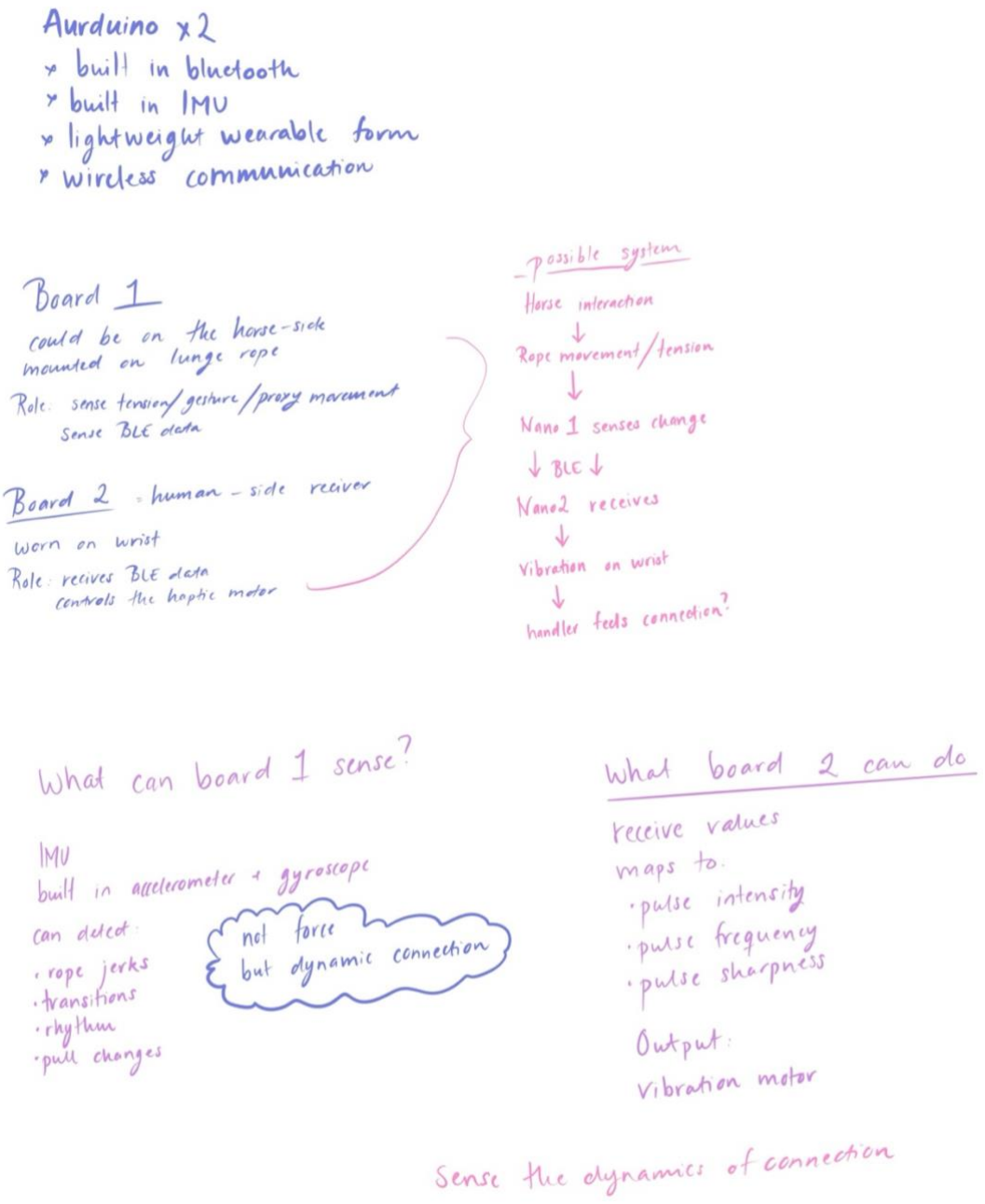


Figure 11. Early exploration of sensing possibilities and system architecture.

Early technical explorations, as seen in Figure 11, investigated what aspects of interaction could potentially be sensed and translated into haptic feedback. Rather than measuring force directly, these explorations focused on dynamic qualities such as rope movement, rhythm, transitions and changes in interaction patterns.

Low-fidelity experimentation was carried out using physical computing components including two Arduino Nano 33 Sense boards, a vibration motor, a potentiometer and a 3D printed wearable structure that could be modified throughout the design process, see figure 12. Arduino Nano 33 Sense boards were selected because they offered built-in sensing capabilities, wireless communication possibilities and a lightweight from suitable for

wearable interaction. Rather than constructing a finalized technological system, the prototypes were intentionally designed to support continuous adaptation and experimentation.



Figure 12. Wearable haptic prototype containing Arduino components and vibration motor.

The wearable prototype was positioned on the wrist to investigate whether haptic feedback could remain integrated into bodily action during lunging. The placement was intended to allow feedback to be perceived without requiring participants to redirect visual attention away from the horse and interaction itself.

The development of haptic feedback patterns evolved through iterative sketching and reflection. Initial explorations considered several possible mappings between interaction qualities observed during fieldwork and different forms of haptic feedback. Rather than functioning as objective representations of horse behaviour, these mappings were used as exploratory tools for investigating how changing interaction qualities could potentially be translated into bodily experiences.

Rope Quality	Haptic Pattern	Meaning
Stable contact	Even Pulse	Sustained connection
Slack/disengagement	Fading Pulse	Reduced connection
Sudden Change	Sharp pulse	Interruption
Rhythmic movement	Rhythmic pulse	Movement Synchrony

Table 3. Early explorations of mappings between interaction qualities and haptic feedback patterns

Initial explorations included several possible interaction states and vibration characteristics, see Table 3, such as pulse timing, rhythm and intensity. However, having four states also increased interpretation demands and complexity.

Rope Quality	Haptic Pattern	Meaning
No contact	No vibration	No connection
Unstable connection	Irregular pulse	searching for connection
Rhythmic movement	Rhythmic pulse	Movement Synchrony

Table 4. Iterative refinement toward simplified haptic interaction states.

The later prototype focused on three states, rather than functioning as objective classifications, these states were intended as exploratory representations of changing interaction qualities.

Because interaction qualities such as connection and attentiveness were difficult to computationally define and detect at this stage, Wizard-of-Oz methods were used before implementing sensing functionality. Within interaction design, low-fidelity and Wizard-of-Oz approaches are commonly used to iteratively explore interaction concepts before complete implementation (Sharp et al., 2019). In this project, this was particularly relevant because the

aim was not to detect objective measurements, but to investigate how changing interaction qualities could be experienced through haptic feedback during horse-human interaction.

During testing sessions, the researcher manually triggered vibration patterns based on interpretations of changing interaction qualities during horse-human interaction. This made it possible to rapidly investigate different forms of haptic experiences without requiring technically complete sensing systems.

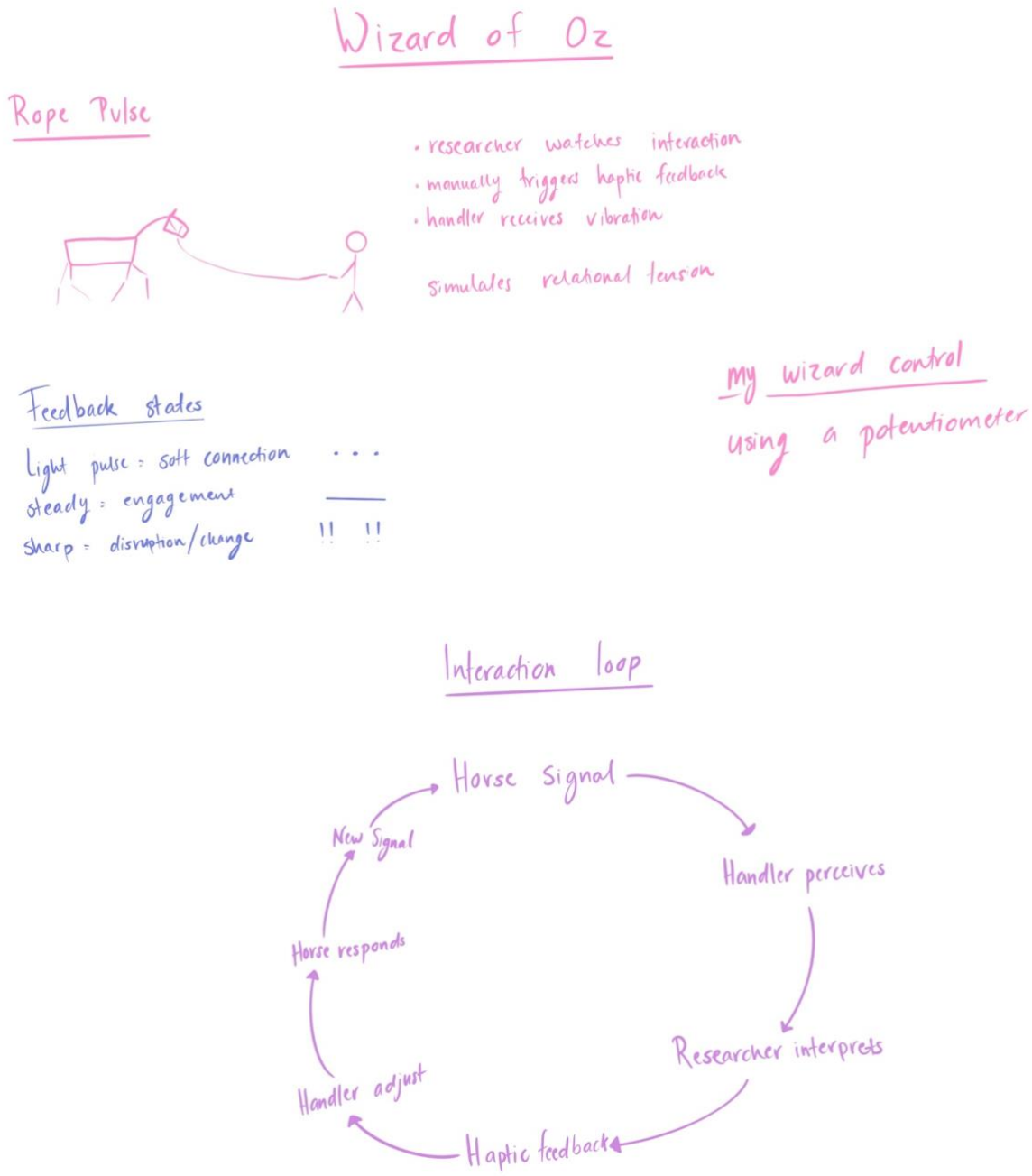


Figure 13. Sketch of the Wizard-of-Oz setup and interaction process.

The Wizard-of-Oz setup, as seen in Figure 13, allowed rapid modification of vibration timing, rhythm and intensity during testing sessions. Early testing session explored several interaction states and their associated haptic patterns. These states were informed by recurring themes identifies during observations, interviews and video analysis and translated into different forms of haptic feedback.

Interaction state	**Observation qualities associated with interaction**	**Haptic pattern**	**Intended experience**
State 1	Reduced attentiveness, distraction, delayed responsiveness, lack of engagement	No vibration	Absence of feedback may encourage reflection on the current interaction
State 2	Inconsistent rhythm, fluctuating responsiveness, changing attention	Irregular and unstable vibration pattern	Draw attention to changes or instability in the interaction
State 3	Stable rhythm, responsiveness, attentiveness, ears focused on handler	Stable and rhythmic vibration pattern	Reinforce awareness of ongoing connection

Table 5. Interaction states refined through Wizard-of-Oz testing and reflection.

During the first testing sessions, reflection on the interaction states led to changes in how connection was understood and represented. Initial explorations focused more strongly on rope tension and changes in the lunge line as potential indicators of connection. The assumption was that changes in tension could be interpreted as changes in interaction quality and later translated into haptic feedback.

However, as the testing sessions progressed, it became increasingly apparent that rope tension alone did not sufficiently represent the qualities participants associated with connection. Moments described as connected interaction were not only related to physical tension in the line, but were instead interpreted through a combination of attentiveness, responsiveness, rhythm and engagement.

This led to a shift in perspective during the design process, see Table 5. Rather than treating rope tension as a direct representation of connection, it became understood as one possible aspect among several relational qualities emerging within the interaction. The design focus therefore shifted away from attempting to objectively detect and judge connection toward exploring more interpretive and relational interaction states.

The testing sessions also resulted in practical adjustments to the haptic feedback itself. Participants initially experience some vibration patterns as difficult to perceive during

movement and environmental distractions. To improve distinguishability during use, vibration intensity was increased and clearer differences between haptic states were introduced.



Figure 14. Participants using the wearable prototype during situated Wizard-of-Oz testing.

The situated nature of the testing was important because interaction qualities such as attention, responsiveness and rhythm could only be meaningfully explored while participants were actively engaged with horses. Observing the prototype during ongoing activity also made it possible to investigate whether the feedback remained integrated into the interaction rather than becoming the primary focus of attention.

The Wizard-of-Oz sessions generated observations regarding how participants interpreted the feedback and how it influenced awareness and interaction. These findings are presented in the following section.

4.6 Prototype Evaluation and Findings

The prototypes were evaluated through situated testing sessions conducted during ongoing lunging interactions. Participants used the wearable haptic prototype while observations and reflective conversations were carried out during and after testing sessions. Rather than focusing on technical performance measures, the evaluation explored how participants

experienced and interpreted the haptic feedback and whether it influenced attentiveness, awareness and interaction during lunging.

The evaluation was informed by qualities identified earlier in the fieldwork and analysis process, including attentiveness, responsiveness, rhythm, engagement and participants description of the horse being "with" the handler. These qualities functioned as interpretive points of reference rather than objective measures and were used to reflect upon how the haptic interaction influenced the ongoing horse-human relationship.

During the first Wizard-of-Oz testing, Participant A interacted with the wearable prototype without receiving prior information about the meaning of the vibration patterns. The intention was to observe how the feedback would be interpretated without predefined explanations. During this session, the participant frequently attempted to understand the relationship between the vibration patterns and the ongoing horse-human interaction (see observations in Table 6 & 7).

The participant described becoming increasingly attentive toward both the horse and the feedback itself:

"I concentrate a bit more"

"I wanted to attain more focus and maintain the attention on the horse the whole time"

Earlier findings from fieldwork had identified attentiveness as important quality associated with connection during lunging, where both horse and handler remained engaged and responsive toward one another. The participant's reflections suggested that the haptic feedback encouraged increased awareness of the ongoing interaction and directed attention toward the horse.

At the same time, the participant also described actively searching for meaning in the feedback:

"Yes, I try to make it vibrate"

Initially this created a tension where attention shifted between interpreting the feedback and attending to the horse.

However, during the second testing session, after the participant had been informed about the intended meaning of vibration states, the interaction appeared to change. Rather than focusing on what the feedback represented, the participant increasingly directed attention toward maintaining interaction qualities associated with stable connection

The participant actively attempted to achieve the stable rhythmic vibration pattern and appeared to adjust interaction with the horse accordingly. This became particularly interesting when viewed in relation to qualities identified earlier in the fieldwork. The participant

appeared to focus more strongly on maintaining attentiveness and responsiveness within the interaction while attempting to sustain a more stable rhythm and coordinated movement with the horse.

The participants behaviour also suggested increased willingness to engage, where attention was continuously directed towards maintaining interaction rather than shifting toward distractions or interruptions. Moments associated with stable vibration often appeared closer to earlier descriptions of perceived connection, where the interaction was experienced as more attentive and where the horse was described as “with” the handler.

Finding	Participant quote	Design Implication
The vibrations were difficult to perceive clearly	“Jag kände inte vibrationerna tillräckligt.”	Haptic feedback needs sufficient clarity and intensity to be noticeable during movement and environmental distraction.
Unclear feedback created uncertainty and distracted attention	“Det var lite fokus för jag ville liksom leta efter dem.”	Ambiguous feedback may shift attention toward the technology rather than the horse–human interaction.
The participant tried to interpret the meaning of the vibrations	“Man ville typ försöka förutspå vad det var.”	Haptic systems require interpretable mappings or contextual framing to support meaningful interaction.
The participant understood the feedback as relational rather than technical	“Jag förstod ju att det var någonting kopplat till när jag interagerade med Bettan.”	Participants naturally interpreted the system as related to communication and interaction rather than measurement.
The participant could not identify exactly what the system represented	“Jag kunde liksom inte riktigt avgöra vad man kollade efter.”	The relationship between haptic patterns and interaction states needs to be more legible.
The participant connected the prototype to communication and understanding	“Det är en kommunikationsgrej.”	The prototype was perceived as supporting interaction and interpretation rather than performance optimisation.
The prototype increased attentiveness	“Man kanske blev lite mer medveten.”	Haptic feedback may encourage greater awareness during lunging.
The participant reflected on inattentive lunging practices	“Man kanske inte är lika inlyssnande på dem.”	The prototype may encourage more reflective and attentive interaction practices.
The vibrations were experienced as natural rather than disruptive	“Det känns ändå naturligt.”	Wearable haptic interaction may integrate well into existing bodily routines
The participant wanted to maintain the connection represented by the vibration	“Man vill ju lite ha kontakten kanske.”	The feedback may motivate attentiveness and relational engagement
Absence of vibration became noticeable	“Det känns lite konstigt när den inte var där.”	Silence/no feedback can become meaningful within relational interaction.
The participant connected lack of vibration to self-reflection	“Man blir ju lite besviken på sig själv.”	The prototype may encourage reflection on one’s own role within the interaction.
The participant believed the system could support understanding of the horse	“Jag tänker att om man har möjlighet att vara mer medveten i att kunna kommunicera så får man nog en gladare häst.”	Participants connected attentiveness and communication to equine welfare and relationship quality.
The participant suggested the system could help with sensitive or reactive horses	“Om det kan vara också preventivt.”	The prototype may have potential in preventative or supportive communication contexts.
Existing strong communication reduced the perceived impact	“Ni hade ju bra kommunikation med varandra.”	The system may become more meaningful in situations where communication is weaker or more uncertain.

Table 7. Observations from first Wizard-of-Oz with participant A

Finding	Participant quote	Design Implication
Clearer and stronger haptic feedback improved understanding	“Det var mycket tydligare vibrationer så då var det mycket lättare för mig att känna skillnaden.”	Haptic feedback needs clear and perceivable patterns in order to support interpretation during interaction.
Understanding the meaning of the feedback changed the experience	“Det var också på något sätt skönare att man kunde förstå vad vibrationerna menade.”	Relational haptic states require some level of onboarding or explanation to become meaningful.
The feedback created immediate awareness of connection	“Nu ser man att det är bättre kontakt och att det blir en direkt feedback på det.”	Real-time haptic feedback may support awareness of interaction quality during lunging.
The participant became more attentive to the horse	“Man ville ha mer fokus och uppmärksamhet på hästen hela tiden.”	Haptics may scaffold attentiveness and encourage sustained focus within the interaction.
Absence of vibration became emotionally meaningful	“Man blev typ besviken.”	Silence/no feedback can communicate reduced connection and function as meaningful relational
The participant actively searched for the vibration states	“Man ville uppnå vibrationerna i form av att man ville ha en bra kontakt.”	The system may encourage intentional engagement and active pursuit of connection.
The prototype reframed lunging as communication rather than passive exercise	“Allt vi gör med hästarna är kommunikation och syfte.”	The prototype may encourage reflection on horse–human interaction and communication
The participant reflected on their own attentiveness	“Man tvingades att vara närvarande på ett annat sätt.”	Haptic interaction may increase bodily and relational awareness during lunging
The participant connected the interaction to care and responsibility	“Bara för att jag har hörlurar i så betyder det ju inte att hon är mindre uppmärksam på mig.”	The prototype may increase awareness of the horse as an active participant in the interaction.
The participant did not want the system in every context	“Jag skulle inte vilja ha det i allt jag gör.”	Haptic interaction may work best as a contextual and occasional support tool rather than continuous feedback.
The participant worried repeated use could reduce attentiveness to the feedback	“Då blir man nog lite så, man märks inte av.”	Overexposure may reduce sensitivity to haptic feedback over time.
Immediate bodily feedback was valued.	"Man kunde få det direkt"	Real-time feedback may support in-the-moment reflection better than delayed analysis

Table 6. Observations from the second Wizard-of-Oz with participant A

Participant B also completed two Wizard-of-Oz testing sessions, see observations in Table 8 & 9. During the first session, the participant received no prior information regarding the intended meanings of the vibration patterns, allowing independent interpretation of the feedback during the interaction.

Similarly to participant A, the participant initially attempted to identify relationships between observable changes in the horse's behaviour and the vibration feedback. During this first session, the participant connected the vibrations directly to visible characteristics of movement:

"It was faster when she went faster"

This interpretation did not correspond to the intended meaning of the vibration states. Rather than representing movement speed, the feedback was intended to communicate changing qualities of interaction. Like participant A, his suggested than when no contextual information was provided, participants actively attempted to create meaning by connecting feedback to visible and measurable aspects of the horse's behaviour.

During the second testing session, after receiving information regarding the intended meaning of the vibration states, the participant increasingly interpreted the feedback in relation to the ongoing horse-human interaction. Rather than focusing on movement speed or isolated cues, attention shifted toward qualities associated with interaction and communication.

The participant described becoming more actively engaged in creating and maintaining connection:

"To try to create contact"

The participant also expressed that the vibrations increased attentiveness during the interaction:

"I concentrate a bit more"

These observations aligned with findings from earlier interviews and field observations, where attentiveness, responsiveness, willingness to engage and rhythmic consistency were repeatedly associated with experiences of connection. Like participant A, the feedback appeared to encourage greater awareness of the ongoing interaction rather than functioning as a system for performance measurement.

There were also occasional moments where the participant interpreted the interaction differently from the Wizard-of-Oz feedback:

"I don't agree with you about the contact sometimes"

Rather than suggesting a failure of the prototype, these moments highlighted that perceived connection is relational and situated rather than fixed or objectively measurable. Earlier observations suggested that experiences of connection emerged through combinations of attentiveness, responsiveness, rhythmic consistency, willingness to engage and reduced signs of tension or distraction. The testing sessions further suggested that these qualities are developed over time through repeated interaction between horse and handler. Because handler build familiarity with their horses through previous experiences, they often develop a deeper and more situated understanding of subtle behavioural changes than an external observer may have. Consequently, what was interpreted as connection by the researcher did not always correspond to how the participant experienced the interaction with their horse.

Finding	Participant quote	Design Implication
The participant noticed changes in vibration rhythm and intensity	"Det blev en ökning av vibrationsmönstren när hon ökade."	Changes in haptic rhythm were noticeable during interaction. But not it's correct implementation
The participant differentiated several vibration states	"Väldigt tysta, lite mer och sen lite längre frekvens..."	Users can distinguish between multiple haptic patterns.
The feedback was interpreted through signal characteristics rather than relational meaning	"Snabbare frekvens och högre frekvens..."	Haptic feedback may initially be understood technically before being interpreted relationally.
The participant did not experience changes in connection with the horse	"Ingenting."	Haptic feedback alone may not immediately alter perceived relational connection.
The participant did not change behaviour toward the horse	"Nej, jag tror inte det."	Awareness effects may require longer exposure or clearer contextual meaning.
Feedback was perceived as supportive when understandable	"Stödjande, om jag vet vad det betyder."	Clear mappings and onboarding are important for meaningful interpretation.

Table 9. Observations from first Wizard-of-Oz with participant B

Finding	Participant quote	Design Implication
The participant and prototype did not always agree about moments of connection	"Jag håller ju inte med dig om kontakten ibland."	Connection is subjective and relational rather than fixed or objectively measurable.
The participant actively searched for the vibration states	"Du försöker ju få det att vibrera."	The feedback encouraged intentional engagement and attentiveness.
The participant focused on creating connection rather than changing behaviour mechanically	"Att försöka skapa en kontakt."	The feedback was interpreted relationally rather than as corrective instruction.
The participant became more attentive toward the horse	"Jag koncentrerar mig lite mer."	Haptic feedback may support attentiveness and ongoing awareness.
The participant suggested value during weak or emerging interaction states	"Då skulle det ju kunna vara ett sätt att hålla mig alert."	The system may be most useful in situations where connection is not yet established.
The participant connected the prototype to reflection on practice	"Det kanske hjälper till lite att uppfatta hästen."	The technology may encourage reflective handling practices.
The participant saw stronger value during rehabilitation or restarting work	"Igångsättning så ser jag det ju på ett annat sätt."	The technology may have stronger value in contexts requiring increased attentiveness and sensitivity.
The participant suggested value for beginners	"Det kan hjälpa nybörjare."	The system may support learning and development of communication skills.
The participant connected the concept to welfare	"Longering är en form av motion... gör du det på fel sätt så skadar du hästen."	Technologies supporting attentiveness may contribute indirectly to equine welfare.
The participant valued immediate embodied feedback	"Jag får ganska så omedelbar respons."	Immediate bodily feedback may support in-the-moment awareness more effectively than delayed analysis.
Delayed feedback through phones was considered less useful	"Jag hade ju inte haft nytta av det om jag behövde gå in i en telefon och kolla sen."	Avoid screen-based and delayed feedback systems.
The participant expressed concern about overreliance on technology	"Man glömmer bort att det är hästar vi håller på med."	Technologies should support rather than replace embodied interpretation and horse knowledge.

Table 8. Observations from second Wizard-of-Oz with participant B.

The findings suggested that the prototype was successful in supporting awareness and attentiveness during lunging rather than functioning as a system for objective measurement. Across both participants, the second testing sessions indicated that the feedback encouraged greater focus on the horse and more active engagement in creating and maintaining connection.

The feedback also appeared to support qualities previously identified during observations and interviews, including responsiveness, attentiveness, rhythmic consistency, willingness to

engage and reduced signs of distraction. Rather than replacing existing bodily interpretation, the haptic feedback functioned as an additional layer of awareness within the ongoing interaction.
At the same time, the evaluation highlighted that connection was experienced as relational and developed over time through repeated interaction between horse and handler. Because participants occasionally interpreted interaction states differently, future iterations may benefit from allowing customization of feedback mappings or sensitivity levels. Rather than assuming a universal definition of connection, the system could adapt to individual horse-handler relationships and communication patterns. Future systems may also provide customization options where handlers could influence how the three haptic states are interpreted or triggered.

Although the Wizard-of-Oz prototype relied on manual interpretation and triggering of haptic states, the testing process also raised questions regarding how such interaction qualities could potentially be identified in future implication. After the testing sessions, different sensing possibilities were explored, focusing more on body dynamics and behavioural cues rather than rope tension. Since the findings suggested that no single measurement appeared sufficient for representing changing interaction qualities. Instead, future systems may potentially rely on combinations of signals interpreted through approaches such as computer vision and AI-based pattern recognition. Rather than identifying connection as a fixed or objective state, such systems could support awareness of changing relational qualities during interaction. These possibilities are explored further in section 6.1.

5. Discussion

5.1 Answering the research questions

This study explored how interaction design could support awareness and interpretation within horse-human communication during lunging, and how real-time technological feedback may influence attention and interaction. Across observations, interviews, design explorations and evaluation activities, the findings suggested that communication during lunging was not experienced through isolated signals or objective measurements. Instead, interaction emerged through combinations of attentiveness, responsiveness, rhythmic consistency, willingness to engage and perceived connection.

The first research question asked:
How can interaction design support awareness and interpretation in horse-human communication during lunging?

The findings suggests that interaction design may support awareness by encouraging sensitivity toward ingoing interaction qualities rather than by providing explicit instructions or objective assessment. Throughout the design process, the project gradually shifted away from attempting to measure connection and instead focused on supporting awareness of changing interaction states. Observations and interviews indicated that handlers relied on multiple subtle cues simultaneously, including rhythm, attention, responsiveness and bodily

expression, rather than single measurable signals. Further explorations of future work can be seen in 6.1.

The second research question asked:
How does real-time technological feedback influence communication, attention and relational awareness between horse and human during lunging?

The Wizard-of-Oz evaluations suggested that real-time haptic feedback primarily influenced participants by increasing attentiveness toward the interaction. During the second testing sessions, after the vibration patterns had been explained, both participants increasingly connected the feedback to relational qualities within the interaction and became more attentive towards their horses. Rather than functioning as direct instruction, the feedback appeared to encourage reflection and active engagement with the ongoing interaction.

Overall, the findings suggest that technology may support horse-human communication not by replacing interpretation, but by acting as an additional layer of embodied support within the existing relationship between horse and handler.

5.2 Embodied and relational interaction

One important finding throughout the project was that horse-human communication during lunging appeared to be embodied and relational rather than easily reduced to objective measurements.

These findings align with Dourish's (2001) perspective on embodied interaction, which argues that meaning emerges through situated bodily activity rather than through detached information processing. Throughout the design process, concepts initially focused on measurable variables such as rope tension or movement characteristics. However, these approaches increasingly appeared insufficient for representing the complexity of horse-human communication. During observations and evaluation sessions, handlers instead relied on subtle bodily and contextual cues including rhythm, posture, attention and responsiveness.

The findings also relate to Haraway's (2008) understanding of human-animal relationships as processes of becoming with, where meaning emerges through ongoing relations rather than existing independently in either participant. Experiences of connection during lunging did not appear to exist as fixed states that could be objectively identified. Instead, they developed through repeated interaction and familiarity between horse and handler. The Wizard-of-Oz testing further supported this understanding, particularly when participants and researcher interpretations occasionally differed regarding moments of connection. Rather than indicating incorrect interpretation, these moments suggested that handlers possess situated knowledge developed through previous experiences with their horses that may not be fully accessible to external observes.

This perspective also aligns with Animal-Computer Interaction research (Mancini, 2011), which argues that animals should not be treated only as passive users of technology but as active participants within interaction design processes. Within this project, the horse was not

approached as an object to be measured or controlled, but as part of an ongoing relationship where interaction emerged between both horse and handler.

Taken together, these findings suggest that technology supporting horse-human interaction may be more meaningful when designed to support awareness and reflection within ongoing interaction rather than attempting to determine or quantify connection itself.

5.3 Support vs Control

A recurring tension throughout the project concerned whether technology should function as a tool for supporting awareness or as a system for evaluation and controlling interaction. Early concepts explored measurable aspects such as rope tension and movement characteristics as indicators for connection. However, observations and prototype testing gradually challenged the idea that interaction could be represented through isolated measurable signals.

Participants rarely described successful interaction through precision or control alone. Instead, qualities such as attentiveness, responsiveness, willingness to engage and the feeling of the horse being “with” the handler were frequently associated with positive interaction experiences. This suggested that horse-human communication was experienced as mutual and relational process rather than a one-directional act of control.

This aligns with More-than-Human design and Animal-Computer Interaction perspectives, where animals are considered active participants rather than passive objects of human action (Mancini, 2011; Wakkary et al., 2015). From this perspective, technology should not primarily strengthen the human’s ability to monitor, correct or control the horse, but instead support awareness and attentiveness within the existing relationship.

During the testing sessions, the haptic feedback appeared to encourage participants to pay closer attention to the interaction itself. Rather than directing behaviour through explicit instruction, the feedback functioned more as a subtle reminder that encouraged reflection and engagement.

The findings therefore suggest that technology within horse-human interaction may become more meaningful when designed to support communication and mutual understanding rather than reinforcing control.

5.4 Limitations

This project has several limitations that should be considered when interpretating the findings. First, the evaluation involved a small number of participants and aimed to generate exploratory insights rather than generalizable results.

The use of Wizard-of-Oz methods also introduced limitations, as vibration patterns were manually controlled and based on the researcher’s interpretation of connection qualities rather than automated sensing. As shown during the evaluation, participants occasionally interpreted

moments of connection differently, highlighting the subjective and relational nature of horse-human interaction.

Finally, the researcher's previous experience with horse handling provided valuable contextual understanding but may also have influenced observations and interpretations throughout the design process.

5.5 Design process

The research-through-design approach supported exploration of an uncertain design space and made it possible for understanding through iterative cycles of observation, prototyping and reflection. Rather than beginning with a predefined solution, the process allowed assumptions to be continuously questioned and reformulated throughout the project.

At the same time, the iterative and exploratory nature of the process also introduced challenges. Because the project moved between observations, synthesis and prototyping, design directions occasionally shifted significantly throughout development. Early concepts focused more on measurable indicators such as rope tension and movement characteristics, which later provided insufficient for representing the relational qualities participants associated with connection. Looking back, earlier reflection on the complexity and subjectivity of connection may have redirected parts of the design process sooner.

The design process was also influenced by the researcher's own experience within horse practice. This provided valuable contextual understanding and enabled closer access to subtle aspects of interaction, but it may also have shaped assumptions regarding what was interpreted as meaningful within the interaction. While observations and participant reflection were used to broaden perspectives, complete separation between prior knowledge and interpretation was difficult.

These shifts and uncertainties also became an important part of the contribution of the project. Rather than identifying a predefined solution, the process generated a gradual shift from measurement and sensing toward awareness, relationality and embodied interaction.

5.6 Future implications

The findings from this project suggest several directions for future development of technology within horse-human interaction. One implication concerns personalization and adaptability. Because experiences of connection appeared to be relational and developed through repeated interaction between horse and handler, future systems may benefit from allowing customization of feedback behaviour rather than relying on predefined interaction states. Features such as adjustable vibration intensity, sensitivity levels or personalized interaction mappings could potentially support different horse-human relationships and practices. This is explored more upon in 6.1 Future Work.

Future work could also investigate how sensing technologies might be implemented while preserving the relational and embodied qualities in this project. Rather than focusing solely on

movement measurement or performance optimization, sensing systems could be explored as ways of supporting awareness and reflection during ongoing interaction.

Finally, longer-term studies involving more participants and horses could provide further understanding of how technological feedback influences interaction over time. This could also create opportunities to investigate how horses themselves respond to technological mediation and whether such systems influence the broader horse-human relationship.

6. Conclusion

This project explored how interaction design can support awareness and interpretation within horse-human communication during lunging and how real-time technological feedback may influence attention and relational awareness. Through a research-through-design approach involving fieldwork, prototyping and Wizard-of-Oz testing, the project investigated how technology could support interaction without disrupting the existing relationship between horse and handler.

The findings suggested that horse-human communication could not easily be represented through isolated signals or objective measurements. Instead, interaction was experiences through changing relational qualities including attentiveness, responsiveness, rhythm and engagement. The design process therefore shifted from exploring measurable indicators toward supporting awareness and embodied interaction.

The project contributes to interaction design by exploring haptic feedback as a way of supporting relational awareness in multispecies context. It contributes practical insights into how technology can be designed to support awareness and interpretation during horse-human interaction without becoming primarily focused on control or performance measurement. The project also contributes to discussions within Animal-Computer Interaction and More-than-Human Design by approaching the horse as an active participant within interaction rather than only an object of observation.

Overall, the findings suggest that technology may have potential in horse-human interaction when designed to support communication, attentiveness and mutual understanding rather than control or optimisation.

6.1 Future work

The Wizard-of-Oz prototype relied on manual interpretation and triggering of haptic feedback. While this allowed interaction qualities to be explored without implementing a complete sensing system, it also raised questions regarding how a future implementation could support similar interpretations automatically.

One direction explored after the evaluation involved body-worn computer vision and AI-assisted interpretation. As discussed in chapters 4 and 5, interaction qualities were not perceived through isolated measurements but through combinations of cues developing

throughout the interaction. Rather than attaching multiple sensors directly to the horse, a future implementation could potentially use cameras integrated into the handler's equipment.

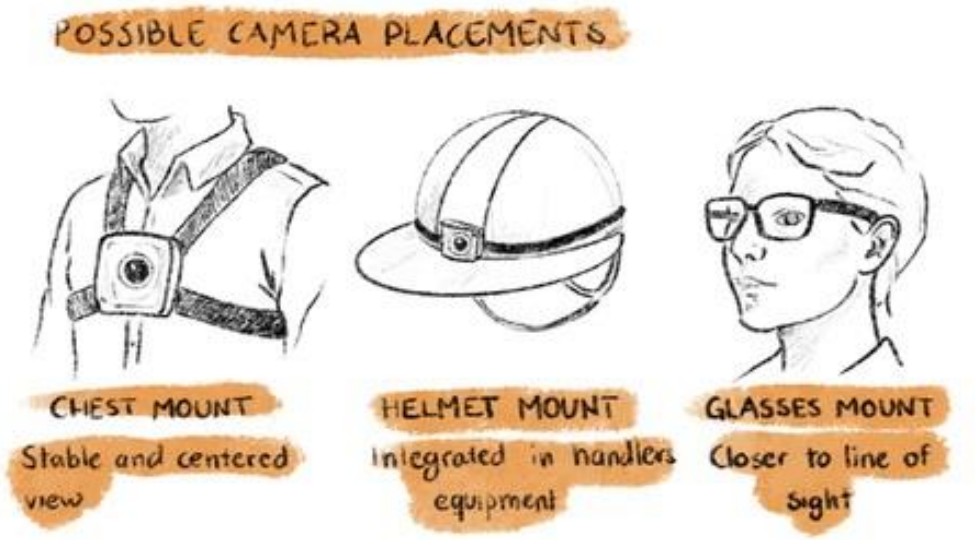


Figure 15. Possible camera placements for body-worn sensing during horse-human interaction. Different placements may provide different perspectives while remaining integrated into existing handling practices.

Possible placements include chest-mounted, helmet-mounted or glasses-mounted cameras. A body-worn approach may allow sensing to remain integrated into existing practices while following the handler's perspectives during lunging.

Using computer vision approaches, a future system could potentially identify combinations of observable behavioural cues during interaction.

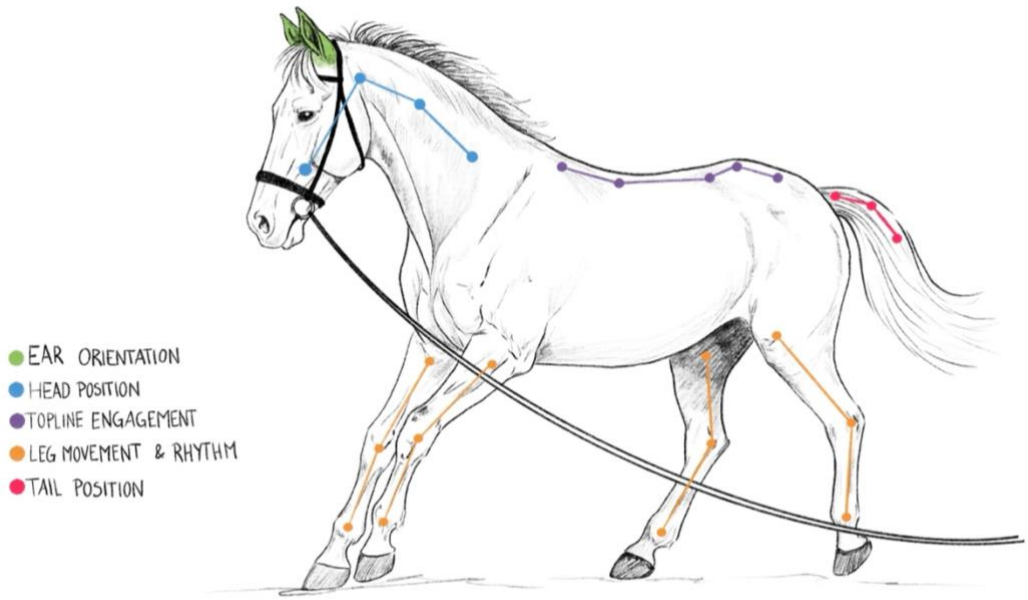


Figure 16. Exploratory visualization of behavioral cues that may potentially be identified through computer vision approaches, including ear orientation, head position, posture, movement rhythm and body dynamics. See bigger version in Appendix B.

These cues would not function as isolated indicators of connection but could instead provide contextual information regarding changing interaction qualities. For example, changes in movement rhythm combined with attentional shifts and posture changes may indicate changing interaction dynamics.

The captured information could potentially be processed through computer vision and AI-based analysis methods.

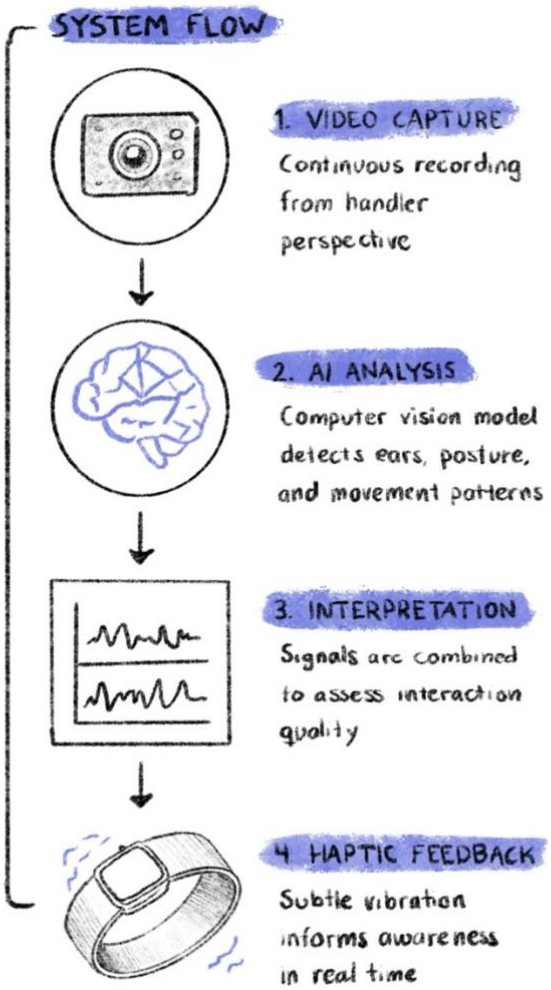


Figure 17. Conceptual system flow illustrating visual capture, AI analysis, interpretation and haptic feedback generation.

Rather than identifying whether connection objectively exists, multiple signals could be interpreted together to identify changing interaction patterns and relational qualities.

The resulting interpretations could then be translated into the simplified haptic states explored throughout this project. Rather than requiring a dedicated wearable device, a future implementation could potentially integrate haptic feedback into existing technologies such as smart watches. Smart watches already contain vibration motors, wireless communication

capabilities and wearable interaction features that may allow feedback to remain discreet and integrated into existing practices without introducing additional equipment.

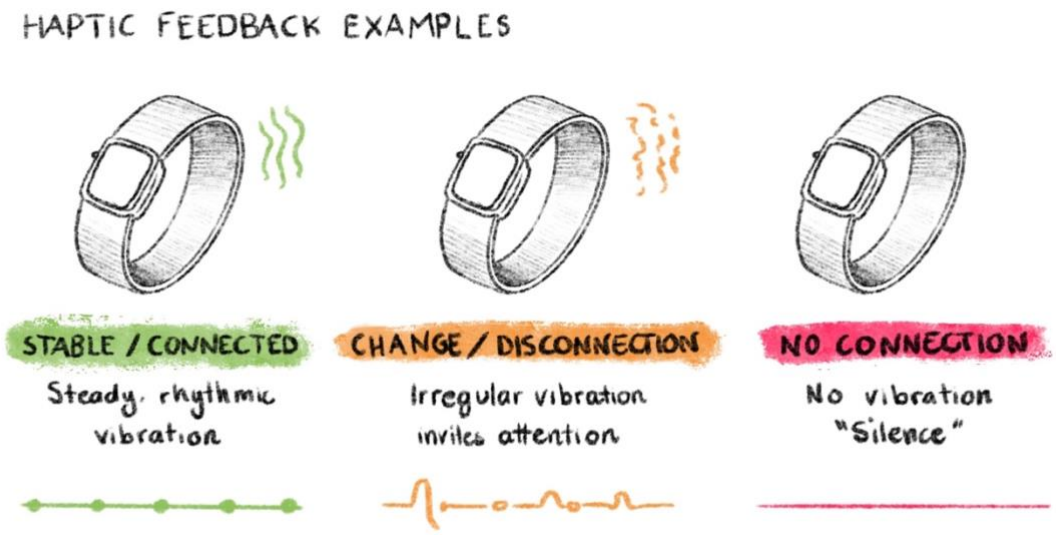


Figure 18. The three states explored throughout this project.

The system could further be understood as integrated part of ongoing interaction rather than a separate technological device where sensing and feedback remain embedded within existing equipment and practices.

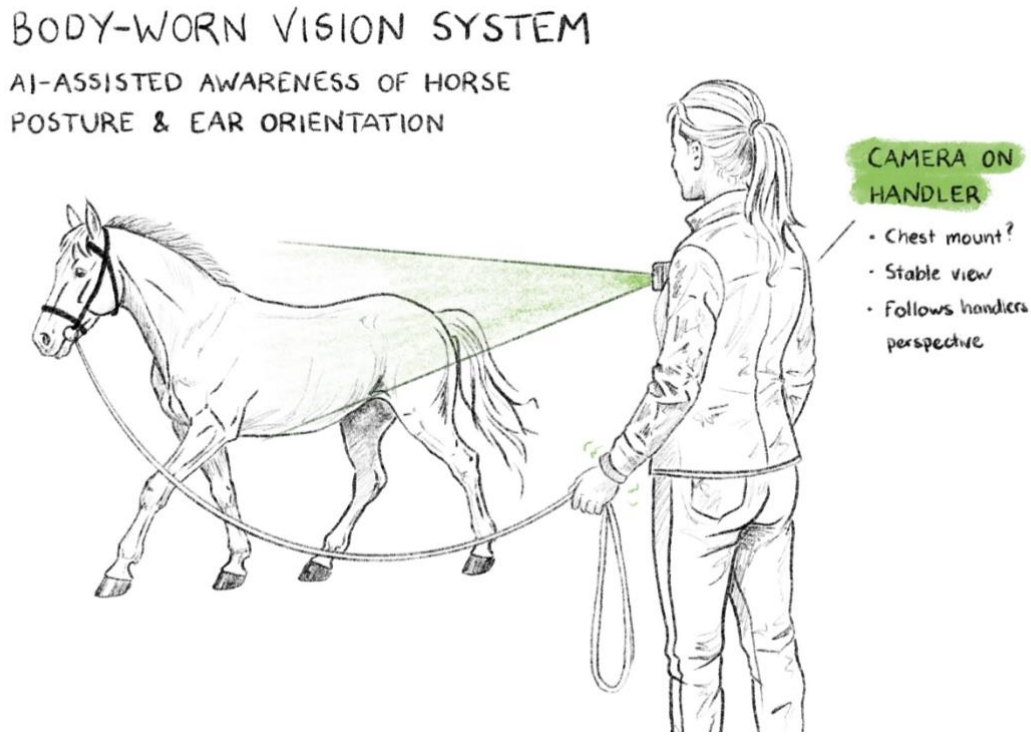


Figure 19. Body-worn vision system concept illustrating camera placement and real-time haptic feedback during lunging interaction. See bigger version in Appendix B.

The evaluation also suggested that future implementations may benefit from customization. Participants occasionally interpreted interaction differently and expressed different preferences regarding feedback. Rather than assuming a universal understanding of interaction, future systems could potentially allow handlers to adjust feedback sensitivity or select which interaction qualities should trigger haptic responses. Over time, systems may also adapt to individual horse-handler relationships and recurring interaction patterns.

To further explore how a personalized system could function in practice, a user journey was developed (Figure 20). The journey illustrates how changing interaction qualities during a lunging session may influence haptic feedback over time. Rather than directing behaviour or controlling the horse, the system is envisioned as supporting awareness during ongoing interaction through subtle, peripheral feedback that allows attention to remain focused on the horse.

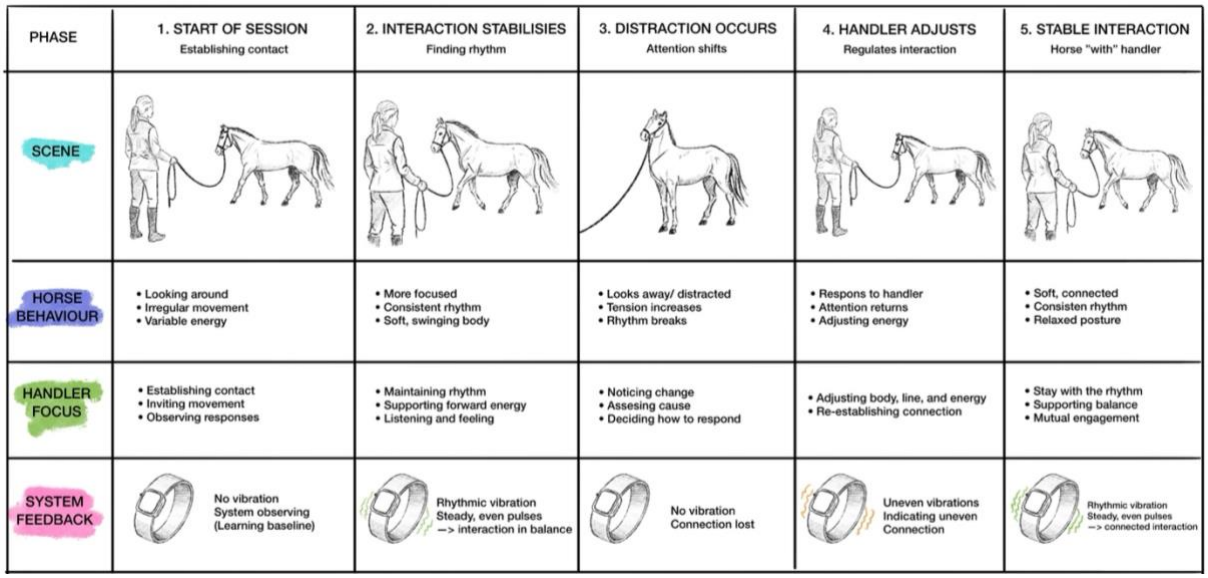


Figure 20. User journey illustrating how changing interaction qualities throughout a lunging session may influence haptic feedback while supporting awareness during horse-human interaction. See bigger version in Appendix B.

These explorations remain speculative and would require investigation regarding both technical feasibility and their influence on horse-human interaction over longer periods of time.

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Appendix A

Interview questions

Interview questions asked in first stages of fieldwork:

1. Can you tell me about your experience with horses?
2. How often do you lunge horses?
3. In what situation do you usually lunge?
4. When you lunge a horse what do you usually pay attention to?
5. How do you know if a horse is moving "well"?
6. What signs tell you that something might be wrong?
7. Are there things that are difficult to see or judge when lunging?
8. Are there situations where you feel unsure?
9. What kind of movement are hardest to interpret?
10. Do you make decisions while lunging, or after?
11. How quickly do you notice small changes in movement?
12. Do you ever miss things and notice them later?
13. How do you feel the horse communicates with you during lunging?
14. Do you notice how the horse reacts to how you move or behave?
15. How do you adjust based in what you notice?
16. Do you use any tools or technology when working with horses?
17. Have you ever used systems that measure movement?
18. What do you think about using technology in this context?
19. If you could get feedback about the horse's movement while lunging, what would be useful?
20. would you prefer to feel it, see it or hear it?
21. Do you think real time feedback would help or be distracting?
22. Do you think technology could improve how we understand horses?
23. Could it also make things worse in any way?

Questions asked during 1st Wizard-of-Oz testing:

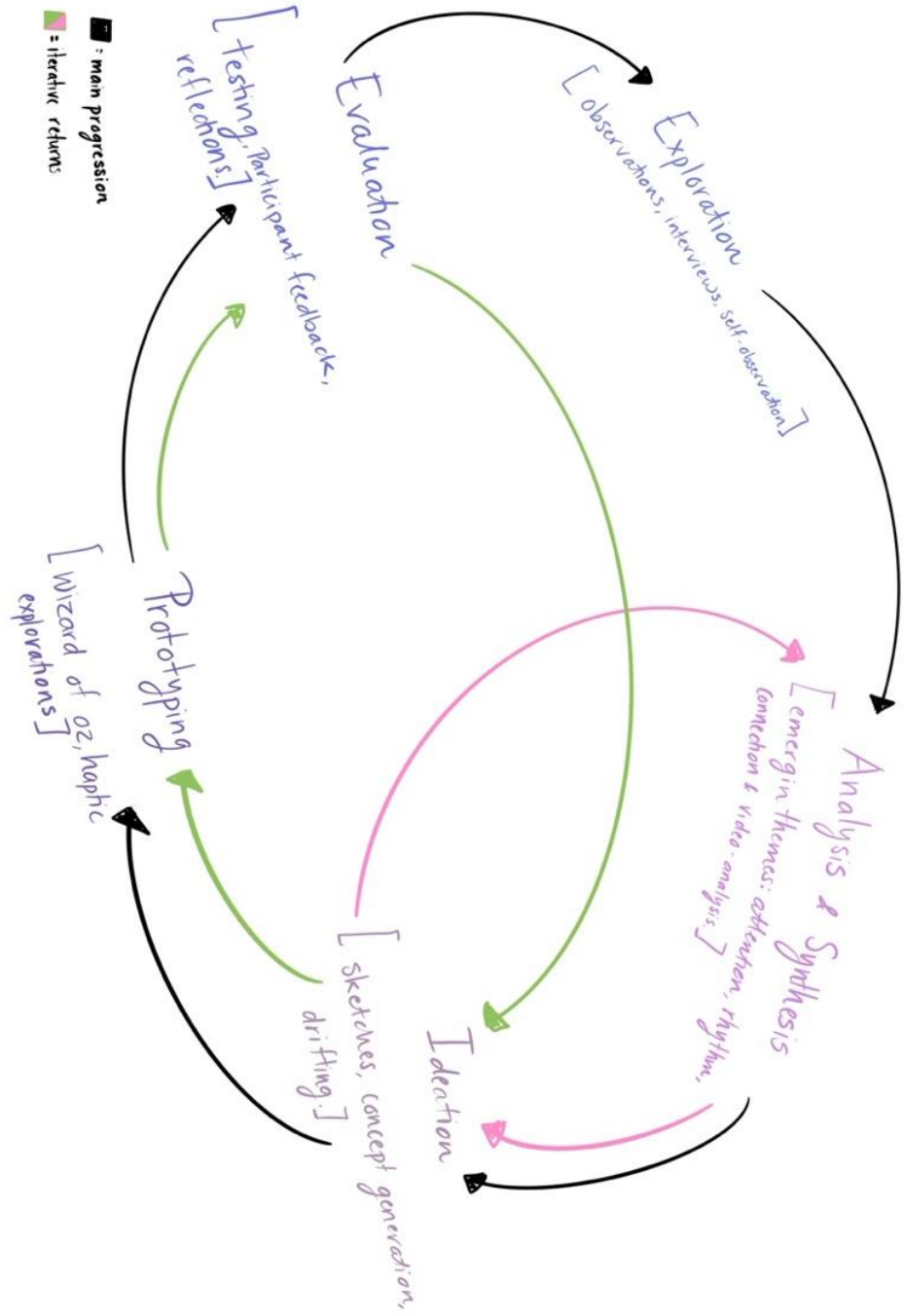
1. What do you usually pay attention to when lunging before using the prototype?
2. Was there anything you noticed differently when using the prototype?
3. How did you perceive or interpret the vibration patterns?
4. Did the prototype affect how you experienced contact or connection with the horse?
5. Did it influence how or when you responded to the horse's signals?
6. How did it feel to receive feedback through vibrations on the body, natural, distracting, or supportive?
7. Do you think the prototype changed the interaction between you and the horse? In what way?
8. Do you think this type of feedback could help you understand the horse better? How?

Observations and questions during 2nd Wizard-of-Oz testing:

1. Do they become calmer?
2. More attentive?
3. More tense?
4. Start “searching” for vibration?
5. Ignore it eventually?
6. “Did it change how present you felt in the interaction?”

Appendix B

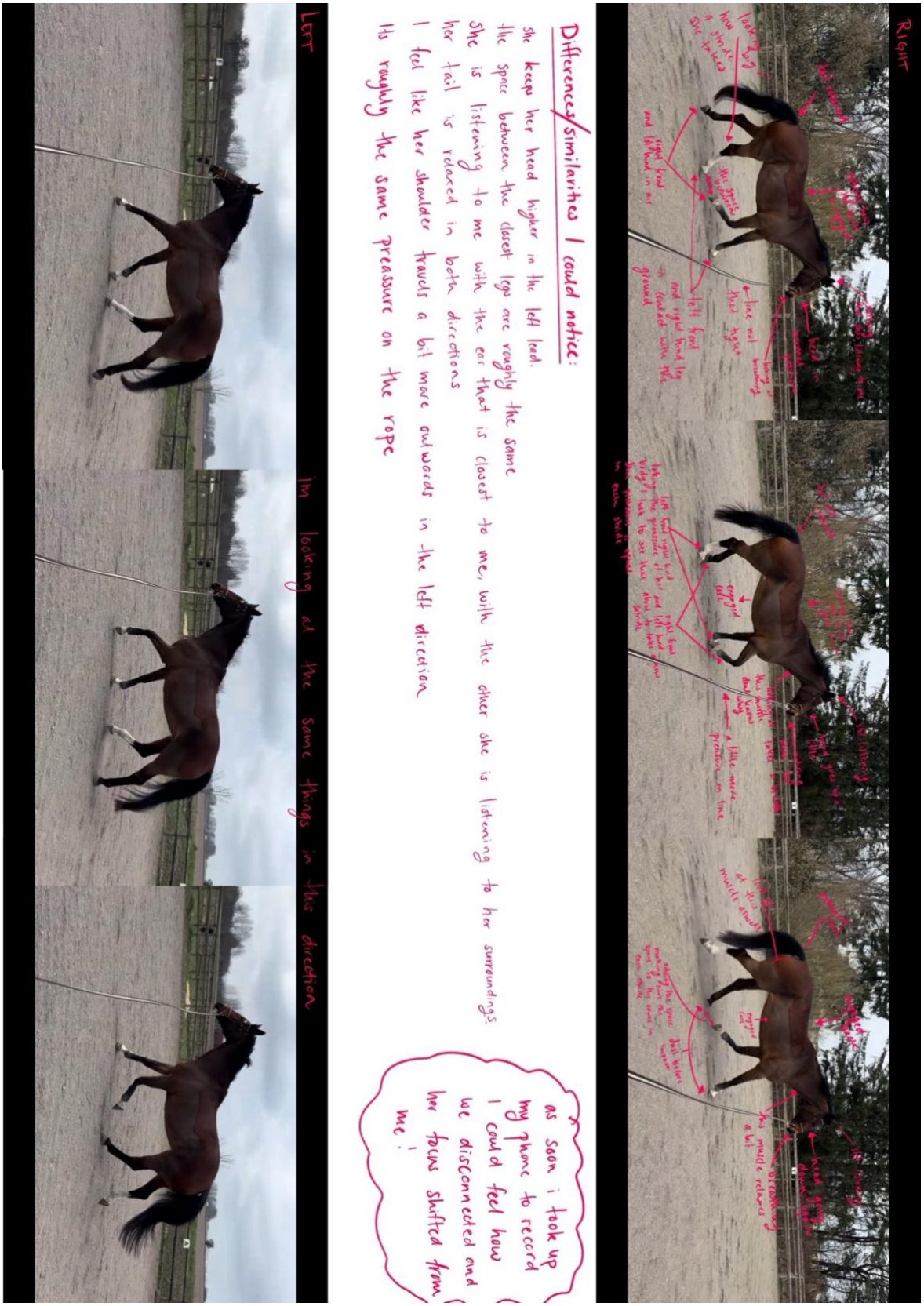
Figures and Tables



Full video analysis from during apprentice-style observation

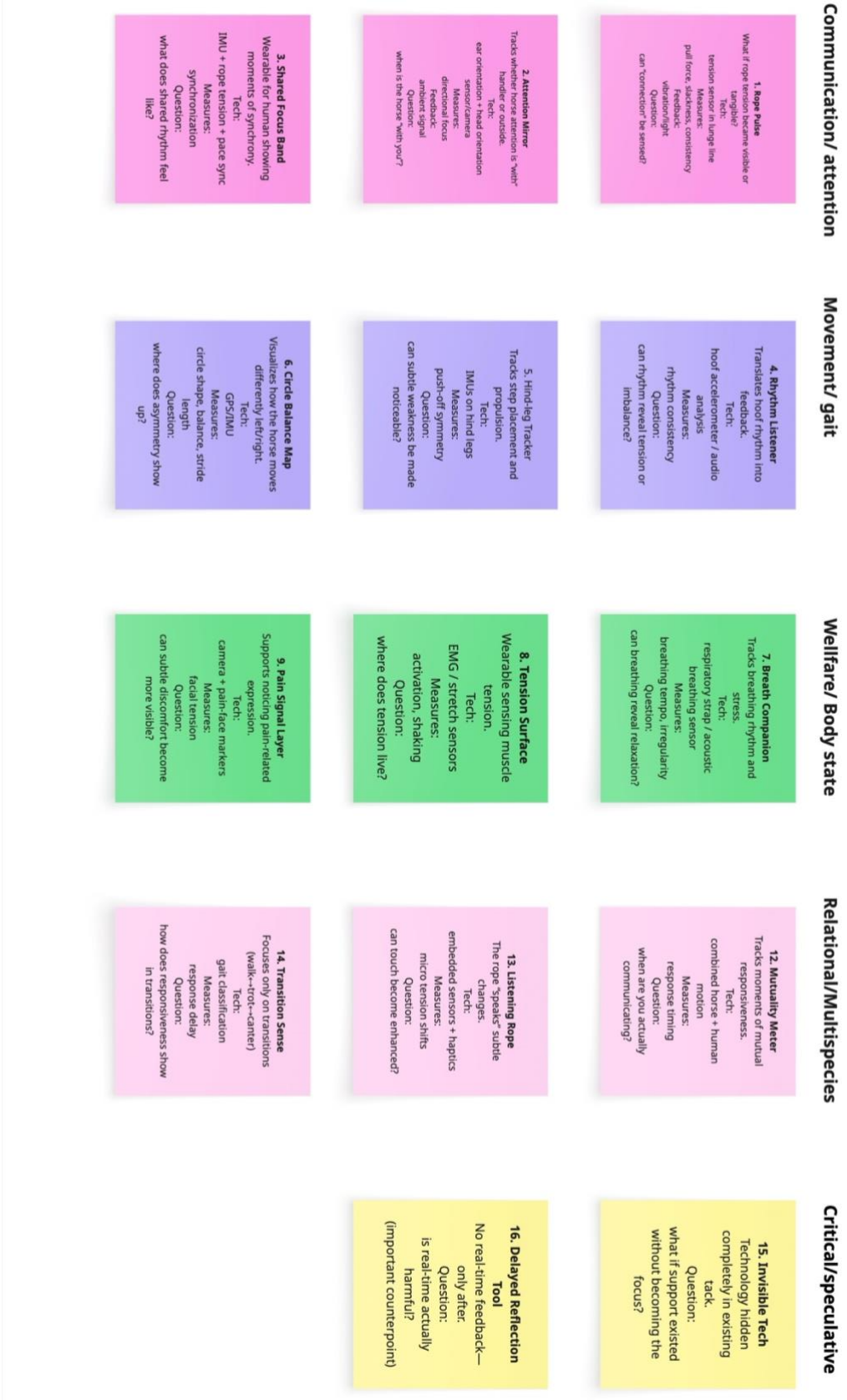


Full video analysis from self-observation



Synthesis of design tensions

Concept exploration



Findings from 1st Wizard-of-Oz session with participant A

Finding	Participant quote	Design Implication
The vibrations were difficult to perceive clearly	"Jag kände inte vibrationerna tillräckligt."	Haptic feedback needs sufficient clarity and intensity to be noticeable during movement and environmental distraction.
Unclear feedback created uncertainty and distracted attention	"Det var lite fokus för jag ville liksom leta efter dem."	Ambiguous feedback may shift attention toward the technology rather than the horse–human interaction.
The participant tried to interpret the meaning of the vibrations	"Man ville typ försöka förutspå vad det var."	Haptic systems require interpretable mappings or contextual framing to support meaningful interaction.
The participant understood the feedback as relational rather than technical	"Jag förstod ju att det var någonting kopplat till när jag interagerade med Bettan."	Participants naturally interpreted the system as related to communication and interaction rather than measurement.
The participant could not identify exactly what the system represented	"Jag kunde liksom inte riktigt avgöra vad man kollade efter."	The relationship between haptic patterns and interaction states needs to be more legible.
The participant connected the prototype to communication and understanding	"Det är en kommunikationsgrej."	The prototype was perceived as supporting interaction and interpretation rather than performance optimisation.
The prototype increased attentiveness	"Man kanske blev lite mer medveten."	Haptic feedback may encourage greater awareness during lunging.
The participant reflected on inattentive lunging practices	"Man kanske inte är lika inlyssnande på dem."	The prototype may encourage more reflective and attentive interaction practices.
The vibrations were experienced as natural rather than disruptive	"Det känns ändå naturligt."	Wearable haptic interaction may integrate well into existing bodily routines
The participant wanted to maintain the connection represented by the vibration	"Man vill ju lite ha kontakten kanske."	The feedback may motivate attentiveness and relational engagement
Absence of vibration became noticeable	"Det känns lite konstigt när den inte var där."	Silence/no feedback can become meaningful within relational interaction.
The participant connected lack of vibration to self-reflection	"Man blir ju lite besviken på sig själv."	The prototype may encourage reflection on one's own role within the interaction.
The participant believed the system could support understanding of the horse	"Jag tänker att om man har möjlighet att vara mer medveten i att kunna kommunicera så får man nog en gladare häst."	Participants connected attentiveness and communication to equine welfare and relationship quality.
The participant suggested the system could help with sensitive or reactive horses	"Om det kan vara också preventivt."	The prototype may have potential in preventative or supportive communication contexts.
Existing strong communication reduced the perceived impact	"Ni hade ju bra kommunikation med varandra."	The system may become more meaningful in situations where communication is weaker or more uncertain.

Findings from 2nd Wizard-of-Oz session with participant A

Finding	Participant quote	Design Implication
Clearer and stronger haptic feedback improved understanding	"Det var mycket tydligare vibrationer så då var det mycket lättare för mig att känna skillnaden."	Haptic feedback needs clear and perceivable patterns in order to support interpretation during interaction.
Understanding the meaning of the feedback changed the experience	"Det var också på något sätt skönare att man kunde förstå vad vibrationerna menade."	Relational haptic states require some level of onboarding or explanation to become meaningful.
The feedback created immediate awareness of connection	"Nu ser man att det är bättre kontakt och att det blir en direkt feedback på det."	Real-time haptic feedback may support awareness of interaction quality during lunging.
The participant became more attentive to the horse	"Man ville ha mer fokus och uppmärksamhet på hästen hela tiden."	Haptics may scaffold attentiveness and encourage sustained focus within the interaction.
Absence of vibration became emotionally meaningful	"Man blev typ besviken."	Silence/no feedback can communicate reduced connection and function as meaningful relational
The participant actively searched for the vibration states	"Man ville uppnå vibrationerna i form av att man ville ha en bra kontakt."	The system may encourage intentional engagement and active pursuit of connection.
The prototype reframed lunging as communication rather than passive exercise	"Allt vi gör med hästarna är kommunikation och syfte."	The prototype may encourage reflection on horse–human interaction and communication
The participant reflected on their own attentiveness	"Man tvingades att vara närvarande på ett annat sätt."	Haptic interaction may increase bodily and relational awareness during lunging
The participant connected the interaction to care and responsibility	"Bara för att jag har hörlurar i så betyder det ju inte att hon är mindre uppmärksam på mig."	The prototype may increase awareness of the horse as an active participant in the interaction.
The participant did not want the system in every context	"Jag skulle inte vilja ha det i allt jag gör."	Haptic interaction may work best as a contextual and occasional support tool rather than continuous feedback.
The participant worried repeated use could reduce attentiveness to the feedback	"Då blir man nog lite så, man märks inte av."	Overexposure may reduce sensitivity to haptic feedback over time.
Immediate bodily feedback was valued.	"Man kunde få det direkt"	Real-time feedback may support in-the-moment reflection better than delayed analysis

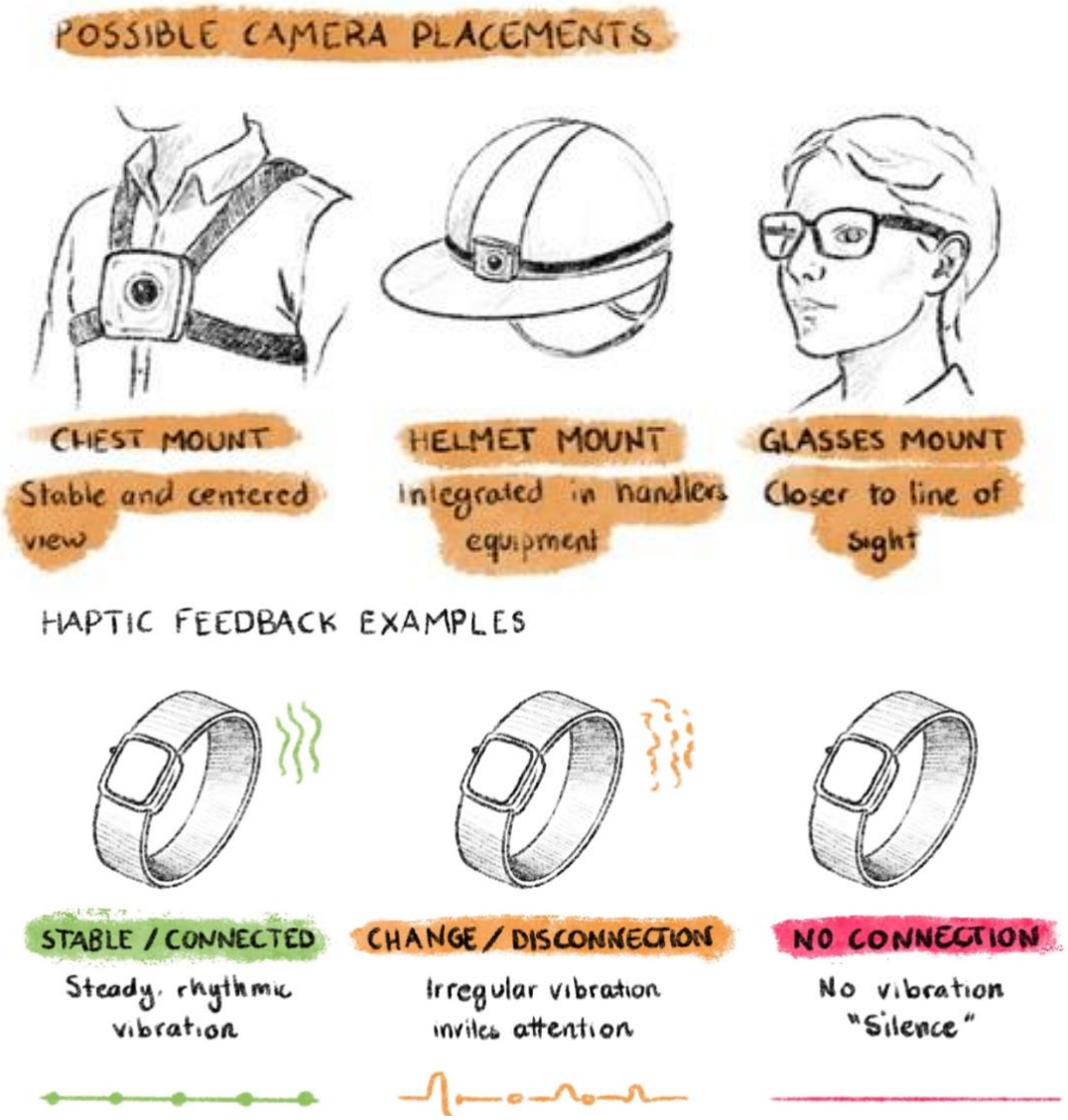
Findings from 1st Wizard-of-Oz session with participant B

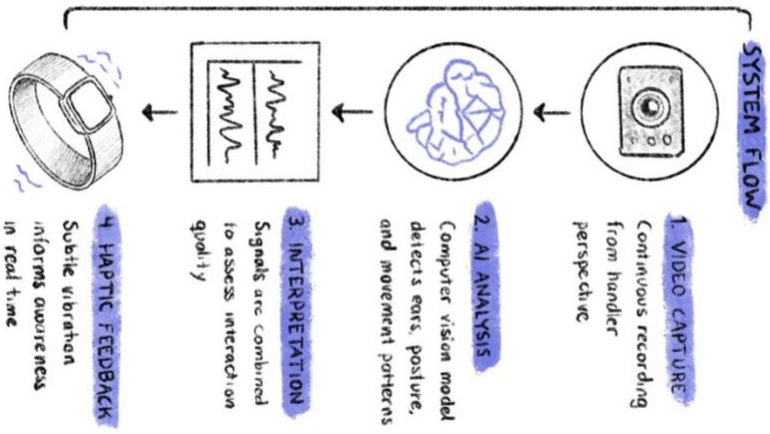
Finding	Participant quote	Design Implication
The participant noticed changes in vibration rhythm and intensity	"Det blev en ökning av vibrationsmönstren när hon ökade."	Changes in haptic rhythm were noticeable during interaction. But not it's correct implementation
The participant differentiated several vibration states	"Väldigt tysta, lite mer och sen lite längre frekvens..."	Users can distinguish between multiple haptic patterns.
The feedback was interpreted through signal characteristics rather than relational meaning	"Snabbare frekvens och högre frekvens..."	Haptic feedback may initially be understood technically before being interpreted relationally.
The participant did not experience changes in connection with the horse	"Ingenting."	Haptic feedback alone may not immediately alter perceived relational connection.
The participant did not change behaviour toward the horse	"Nej, jag tror inte det."	Awareness effects may require longer exposure or clearer contextual meaning.
Feedback was perceived as supportive when understandable	"Stödjande, om jag vet vad det betyder."	Clear mappings and onboarding are important for meaningful interpretation.

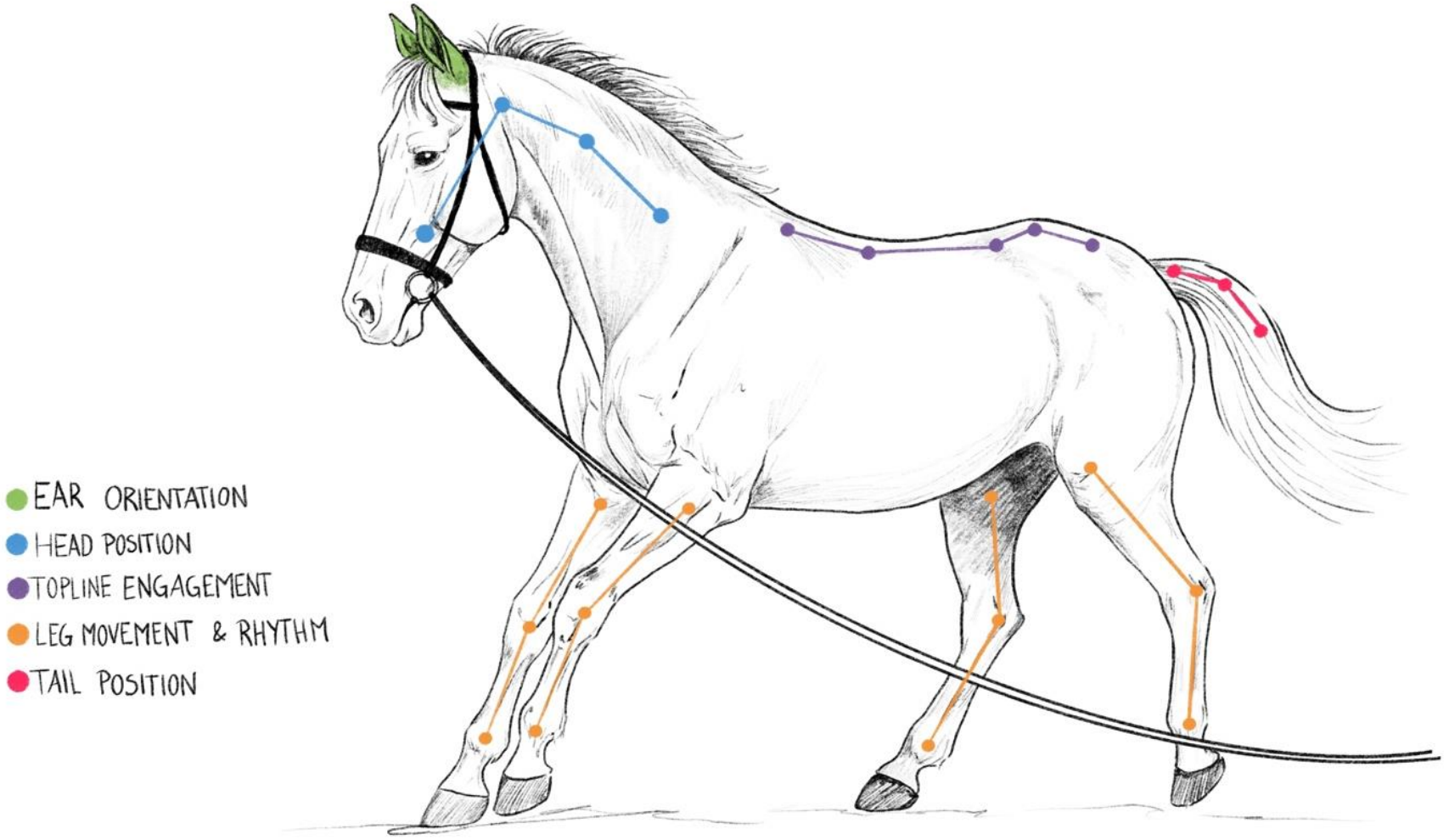
Findings from 2nd Wizard-of-Oz testing session with participant B

Finding	Participant quote	Design Implication
The participant and prototype did not always agree about moments of connection	"Jag håller ju inte med dig om kontakten ibland."	Connection is subjective and relational rather than fixed or objectively measurable.
The participant actively searched for the vibration states	"Du försöker ju få det att vibrera."	The feedback encouraged intentional engagement and attentiveness.
The participant focused on creating connection rather than changing behaviour mechanically	"Att försöka skapa en kontakt."	The feedback was interpreted relationally rather than as corrective instruction.
The participant became more attentive toward the horse	"Jag koncentrerar mig lite mer."	Haptic feedback may support attentiveness and ongoing awareness.
The participant suggested value during weak or emerging interaction states	"Då skulle det ju kunna vara ett sätt att hålla mig alert."	The system may be most useful in situations where connection is not yet established.
The participant connected the prototype to reflection on practice	"Det kanske hjälper till lite att uppfatta hästen."	The technology may encourage reflective handling practices.
The participant saw stronger value during rehabilitation or restarting work	"Igångsättning så ser jag det ju på ett annat sätt."	The technology may have stronger value in contexts requiring increased attentiveness and sensitivity.
The participant suggested value for beginners	"Det kan hjälpa nybörjare."	The system may support learning and development of communication skills.
The participant connected the concept to welfare	"Longering är en form av motion... gör du det på fel sätt så skadar du hästen."	Technologies supporting attentiveness may contribute indirectly to equine welfare.
The participant valued immediate embodied feedback	"Jag får ganska så omedelbar respons."	Immediate bodily feedback may support in-the-moment awareness more effectively than delayed analysis.
Delayed feedback through phones was considered less useful	"Jag hade ju inte haft nytta av det om jag behövde gå in i en telefon och kolla sen."	Avoid screen-based and delayed feedback systems.
The participant expressed concern about overreliance on technology	"Man glömmer bort att det är hästar vi håller på med."	Technologies should support rather than replace embodied interpretation and horse knowledge.

Future Work

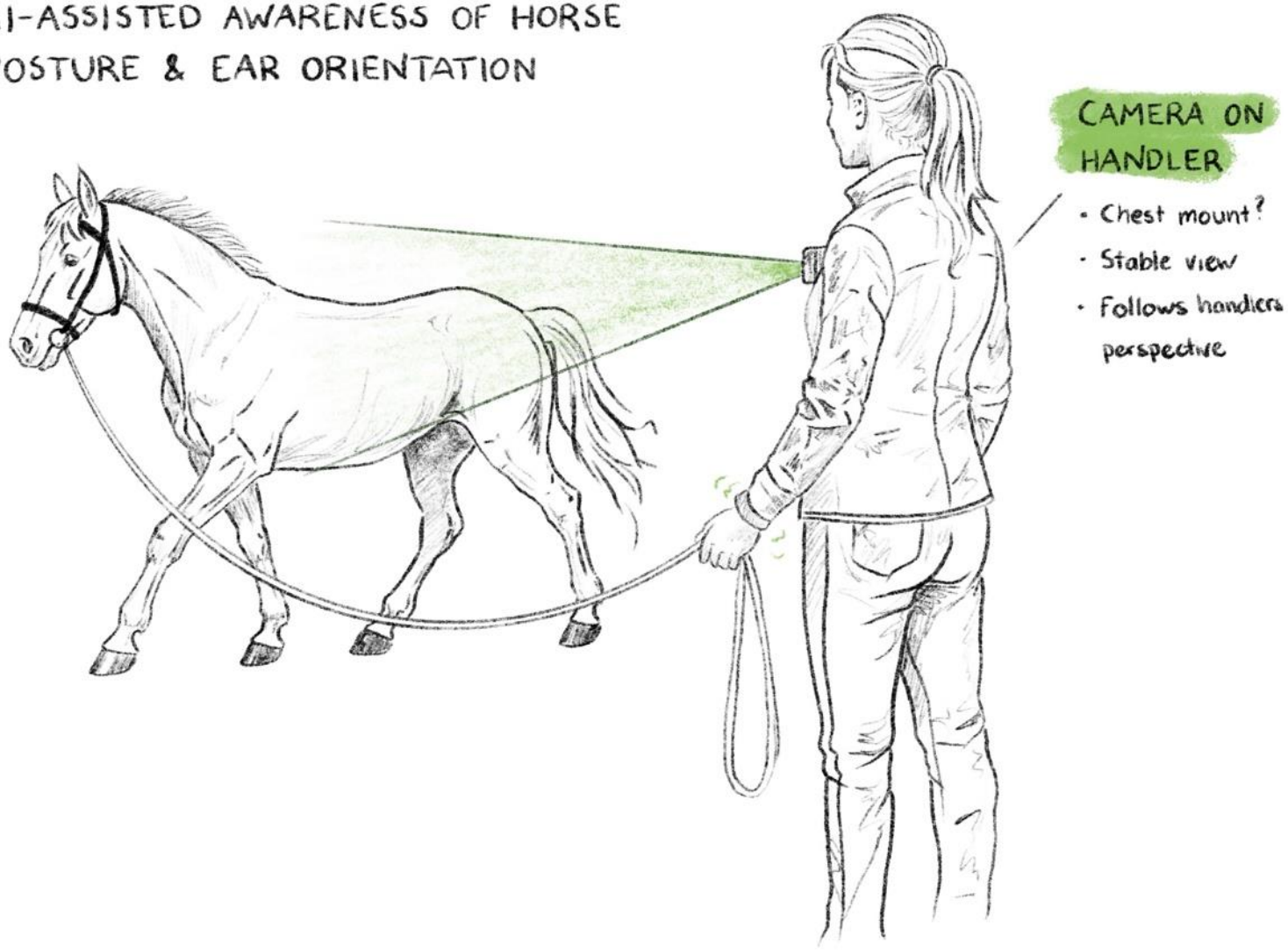

POSSIBLE CAMERA PLACEMENTS
CHEST MOUNT
Stable and centered view
HELMET MOUNT
Integrated in handlers equipment
GLASSES MOUNT
Closer to line of sight
HAPTIC FEEDBACK EXAMPLES
STABLE / CONNECTED
Steady, rhythmic vibration
CHANGE / DISCONNECTION
Irregular vibration invites attention
NO CONNECTION
No vibration "Silence"

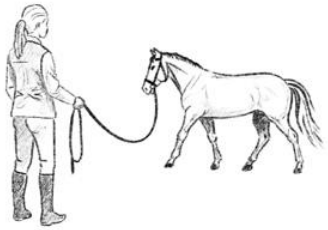
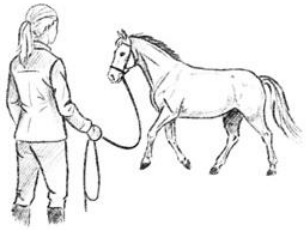
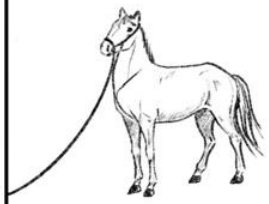
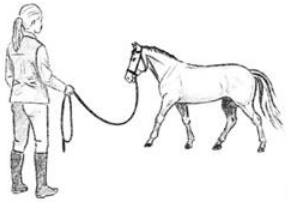
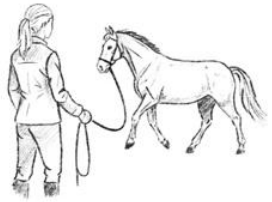

SYSTEM FLOW
1. VIDEO CAPTURE
Continuous recording from handler perspective
2. AI ANALYSIS
Computer vision model detects ears, posture, and movement patterns
3. INTERPRETATION
Signals are combined to assess interaction quality
4. HAPTIC FEEDBACK
Subtle vibration informs awareness in real time


EAR ORIENTATION
HEAD POSITION
TOPLINE ENGAGEMENT
LEG MOVEMENT & RHYTHM
TAIL POSITION

BODY-WORN VISION SYSTEM

AI-ASSISTED AWARENESS OF HORSE POSTURE & EAR ORIENTATION



PHASE	1. START OF SESSION Establishing contact	2. INTERACTION STABILISIES Finding rhythm	3. DISTRACTION OCCURS Attention shifts	4. HANDLER ADJUSTS Regulates interaction	5. STABLE INTERACTION Horse "with" handler
SCENE					
HORSE BEHAVIOUR	• Looking around • Irregular movement • Variable energy	• More focused • Consistent rhythm • Soft, swinging body	• Looks away/ distracted • Tension increases • Rhythm breaks	• Respons to handler • Attention returns • Adjusting energy	• Soft, connected • Consisten rhythm • Relaxed posture
HANDLER FOCUS	• Establishing contact • Inviting movement • Observing responses	• Maintaining rhythm • Supporting forward energy • Listening and feeling	• Noticing change • Assesing cause • Deciding how to respond	• Adjusting body, line, and energy • Re-establishing connection	• Stay with the rhythm • Supporting balance • Mutual engagement
SYSTEM FEEDBACK	 No vibration System observing (Learning baseline)	 Rhythmic vibration Steady, even pulses —> interaction in balance	 No vibration Connection lost	 Uneven vibrations Indicating uneven Connection	 Rhythmic vibration Steady, even pulses —> connected interaction